A factorized Pauli–Villars multigrid method for the Möbius domain-wall operator

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draft

October 22, 2026

Solvers for the Möbius domain-wall operator are slow because its spectrum surrounds the origin. Kanamori, Chen and Matsufuru showed that the preconditioned operator $A' = (D_{\mathrm{PV}} C_{\mathrm{PV}}^{-1})^\dagger D C^{-1}$, built from the Pauli–Villars operator and the site-diagonal part C, has all its eigenvalues in the right half-plane. We construct a two-grid method for A' on the even sites. Its coarse-grid correction is factorized: one coarse solve for the quark operator and one for the adjoint Pauli–Villars operator, each with a nearest-neighbor coarse stencil. The prolongation is built from approximate low modes lifted to the whole lattice, and the correction is exact on these vectors. The smoother is a fixed Chebyshev polynomial. The same multigrid also solves with $D^\dagger$, as HMC requires: the exact relation between D and $D^\dagger$ for Möbius fermions makes $\Gamma_5 D^{-1} \Gamma_5$ a good preconditioner for $D^\dagger$, with no second setup. The method converges although the field of values of the preconditioned operator reaches into the left half-plane; we explain this with its spectrum ($\mathrm{Re}\,\lambda \geq 0.78$) and pseudospectra. On a quenched $24^3 \times 48$ configuration the multigrid iterations grow by a factor of 2–4 from $m = 0.1$ to $m = 0.001$, against 20 for even-odd CGNE. At $m = 0.001$ a solve is 4.5–4.8 times faster than mixed-precision CGNE, and a point-source propagator takes half its time, setup included. The results come from one configuration per volume.

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I. Introduction

Many lattice computations need fermions whose chiral symmetry is broken only slightly at finite lattice spacing. The overlap operator [1, 2] has an exact lattice chiral symmetry. Domain-wall fermions [3, 4, 5] approximate it at a lower cost. They add a fifth dimension of extent L_s, and a large negative Wilson mass $-M_5$ binds modes of opposite chirality to its two boundaries. The residual breaking of chiral symmetry falls as L_s grows. The Möbius generalization [6, 7, 8] adds two coefficients (b, c) and improves the approximation at fixed L_s; $c = 0$ is Shamir's operator.

The price is paid in the linear solver. The domain-wall operator is neither Hermitian nor positive, and one application costs L_s four-dimensional Wilson hops. The standard solver is still the conjugate gradient method on the normal equation of the even-odd operator (CGNE), whose condition number is the square of that of the operator. For Wilson fermions, multigrid solved this problem. The low modes that slow down Krylov solvers are locally coherent [10]: a few of them, cut into blocks, span the low-mode space. Adaptive aggregation multigrid uses this, and its cost hardly depends on the quark mass [11, 12, 13].

Multigrid for domain-wall fermions. For domain-wall fermions multigrid is harder. Because of the large negative Wilson mass, the spectrum encloses the origin. Non-Hermitian Krylov methods then converge only after a number of steps of the order of the system size [15, 16]. Boyle [16] lists four approaches that work to some extent. They differ in the operator that is coarsened.

- Cohen *et al.* [17] coarsened the squared operator, with blocks that cover the whole fifth dimension. Its coarse operator reaches two hops.
- HDCG [18] accelerates CGNE and coarsens the squared even-odd Schur complement. Its coarse stencil has 81 points for blocks of at least 4^4.
- HDCR [19] coarsens the Hermitian indefinite operator $\gamma_5 R_5 D$ at $c = 0$. Its coarse operator stays nearest-neighbor, which keeps the setup affordable in HMC.
- Brower, Clark, Howarth and Weinberg [21] noted that the Pauli–Villars preconditioned operator $D_{\mathrm{PV}}^{-1} D$ carries the physical four-dimensional degrees of freedom. They solved $D_{\mathrm{PV}}^{\dagger} D x = D_{\mathrm{PV}}^{\dagger} b$ with coarse operators from the projected Wilson kernel, in the Schwinger model.

Boyle and Yamaguchi [20] compared these approaches for Shamir fermions; on large volumes HDCR was set up and solved faster than a single even-odd CG solve. Two of their findings matter here. First, CG on the Schur complement $M_{ee} - M_{eo} M_{oo}^{-1} M_{oe}$ needed far fewer iterations than on the left-preconditioned $1 - M_{ee}^{-1} M_{eo} M_{oo}^{-1} M_{oe}$. Second, an even-odd smoother did not work with the HDCR correction; they traced this to eigenvectors that are not aligned.

Recent work by Boyle and collaborators. The multiple right-hand-side HDCG of Ref. [16] solves the normal equation of the even-odd operator with a two-level conjugate gradient. Its coarse space is spanned by blocked low modes from Chebyshev filters. Treating many right-hand sides at once makes the coarse space cheap, and gains of up to twenty over even-odd CG are reported at physical quark mass. In a Lattice 2026 talk, Boyle and Oare [22] named two limits of this approach: forming the 81-point coarse operator is expensive, and a single solve on a new gauge field, as in HMC, gains nothing from many right-hand sides. They coarsen the two-hop operator $D_{\mathrm{PV}}^{\dagger} D$ instead, which gives a 33-point stencil. They stressed that $D_{\mathrm{PV}}^{\dagger} D$ is non-normal: its field of values can reach the left half-plane although its spectrum does not, so the GCR bound of Eisenstat, Elman and Schultz [23] is lost. They added a shift to the smoothers.

Setup cost in ensemble generation. A multigrid setup is paid once per gauge field. This is affordable for propagators but not for HMC, where the field changes at every step. Eick and Mawhinney [34] avoid the setup by renormalization-group blocking: a fine Möbius ensemble is blocked to a coarse lattice whose domain-wall operator approximates the low modes of the fine one. They find that the interpolated correction carries high modes that the fine operator amplifies by four orders of magnitude, and they need a Chebyshev low-pass filter of degree 150–300.

The preconditioned operator. Kanamori, Chen and Matsufuru [24] examined the spectra of several preconditioned Möbius operators. The operator $A' = (D_{\mathrm{PV}} C_{\mathrm{PV}}^{-1})^{\dagger} D C^{-1}$, which combines the Pauli–Villars factor with the site-diagonal part C, has all eigenvalues in the right half-plane and the smallest condition number among those they studied. With D_{PV} truncated to blocks, negative real parts come back.

Separately, Neff's form of the domain-wall matrix [25, 26] rescales the fifth dimension by a site-local factor Q and changes the conditioning of the even-odd system.

This note. We construct a two-grid method for A'. It is an adaptive aggregation multigrid in the sense of Refs. [11, 12, 13]: the coarse space comes from computed near-null vectors cut into aggregates, and the coarse operators are Galerkin projections. What is new is the coarse-grid correction. The outer system is the even-odd one, as in HDCG. But instead of coarsening the even-odd operator, we replace the inverses of its two factors, the quark operator and the adjoint Pauli–Villars operator, by two separate Galerkin inverses. Each has a nearest-neighbor coarse stencil. The prolongation is built from approximate low modes lifted to the whole lattice in two ways, so that the correction is exact on them. The smoother is a fixed Chebyshev polynomial, and the setup combines Chebyshev filtering with a bootstrap. For HMC we also solve with $D^\dagger$. We show that the multigrid of D, placed between two Γ_5, is a good preconditioner for $D^\dagger$, with no second setup. We give numerical results: the field of values, spectrum and pseudospectra of the preconditioned operator, the behavior of truncated GCR, a mass scan and point-source propagators against CGNE, the sensitivity to the parameters, and the solves with $D^\dagger$.

Section II describes the method in words. Section III recalls the spectra the construction rests on. Sections IV–X give the method: the operator and its preconditioning, the even-odd forms, the factorized correction, the coarse solvers, the setup, the smoother and the outer solver. Section XI solves with $D^\dagger$. Section XII gives the results: first why the method converges, then its cost. Section XIII lists open problems. Appendix A relates our notation to other references, and Appendix B describes the implementation and how to reproduce the results.

Conventions. Five-dimensional fields are $\psi_s(x)$ with $s = 0, \dots, L_s - 1$, and $P_+, P_- = \frac{1}{2}(1 \pm \gamma_5)$. R reverses the fifth dimension, $(R\psi)_s = \psi_{L_s-1-s}$, and $\Gamma_5 = \gamma_5 R$. The subscripts e and o denote blocks on the even and odd four-dimensional sites, and X_{pq} maps parity q to parity p. The injection ι puts a field on the even sites onto the whole lattice, with zero on the odd sites, and $\pi = \iota^\dagger$ takes the even sites. Section IV defines the operator and its parts C and Q.

II. Overview of the method

This section describes the method in words, for readers who know domain-wall fermions but not multigrid. The later sections give the equations.

Multigrid in one paragraph. A multigrid preconditioner removes the error of an iterate in two parts. The high modes are damped by a few steps of a cheap iteration on the fine lattice, the *smoother*. The low modes, the near-zero modes that make CG slow, are removed by solving a much smaller problem on a *coarse lattice*. The coarse lattice comes from *aggregation*. We cut the fine lattice into blocks of 4^4 sites that contain the whole fifth dimension. Each block becomes one coarse site, and its degrees of freedom are a few approximate low modes restricted to the block, the *test vectors*. The *prolongation* $\mathcal{P}$ maps a coarse vector to the fine lattice through these vectors, and $\mathcal{P}^\dagger$ maps back. The coarse operator of a fine operator X is its *Galerkin* projection $\mathcal{P}^\dagger X \mathcal{P}$; it couples only neighboring coarse sites if X couples only neighboring fine sites. A few vectors per block suffice because low modes are locally coherent [10]. Finding the test vectors is the *setup*. It is paid once per gauge field and shared by all right-hand sides.

What is solved. As in even-odd CGNE, we reduce $D\psi = \eta$ to the even sites, here with the preconditioning of A'. This gives a system $\tilde{A}y = \rho$, and we recover ψ from y afterwards. In practice $\tilde{A}$ is the left form $\tilde{A}_L$, which can carry Neff's α (Sec. V). The outer solver is flexible GCR that keeps two directions, GCR(2). Each of its iterations applies one two-grid cycle as the preconditioner (Sec. X).

One cycle. A cycle is the smoother (a Chebyshev polynomial in $\tilde{A}$), a coarse correction, and the smoother again (Algorithm 3). The coarse correction is the new part. Coarsening $\tilde{A}$ itself would give a coarse stencil of 81 points, because $\tilde{A}$ reaches four hops on the even sites. Instead we invert its two factors separately:

first the adjoint Pauli–Villars operator, then the quark operator. Each inverse is replaced by the Galerkin inverse of D_{PV} or D, whose coarse stencil has 9 points (Fig. 4, Sec. VI). Both coarse systems are solved only roughly, to a relative residual of about 0.1, in single precision and on the even coarse sites (Sec. VII).

The setup. The setup has four steps (Algorithm 2, Sec. VIII).

1. A Chebyshev filter applied to random vectors gives approximate low modes y of $\tilde{A}_R$: the vectors of its smallest singular values.

2. A bootstrap improves them. We build the cycle from the current vectors and apply a few FGMRES steps with it to each vector; this enriches the vector in what the cycle still misses.

3. We lift each y to the whole lattice in two ways, s and $\hat{t}$, one for each coarse solve. This makes the correction exact on these vectors (Proposition 1).

4. We split the lifts by their Γ_5-chirality and orthonormalize them on each block. This gives $\mathcal{P}$, and from it the coarse operators of D and D_{PV}.

One setup serves both forms and every α.

Solving with $D^\dagger$. HMC also solves with $D^\dagger$. The same cycle, placed between two Γ_5, is a good preconditioner for it (Sec. XI), so no second setup is needed.

Where to read on. The construction is in Secs. V and VI; Table 1 compares it with other multigrid methods for domain-wall fermions. The parameters used in practice are in Table 4, and the comparison with CGNE is in Sec. XII.F.

III. Spectra of preconditioned Möbius operators

Where the spectrum lies decides which iterative methods can work. A Krylov method builds a residual polynomial p with $p(0) = 1$ that should be small on the spectrum. Such a polynomial of low degree exists only if a region that excludes the origin, such as an ellipse, encloses the spectrum. The Chebyshev smoother of Sec. IX needs the same, since it is defined by an ellipse in the right half-plane. In short, Krylov solvers and polynomial smoothers work when the eigenvalues lie on one side of the origin. Those of the domain-wall operator do not, and the preconditioners below move them there. Figures 1–3, from Ref. [24], show what each preconditioner does; the captions give the lattices.

The unpreconditioned operator. The free spectrum (Fig. 1) is a band that reaches into $\mathrm{Re}\,\lambda < 0$. On a $2+1$-flavor configuration at the physical point (Fig. 2, top left) the real parts lie in $[-1.29, 13.2]$. So CG and BiCGStab do not work on D itself, and the practical choice has been CG on a positive definite squared operator.

Site-diagonal preconditioning. The site-diagonal part C has an inverse that is the same matrix on every site (Sec. IV.C). With it, DC^{-1} has real parts in $[-0.22, 2.22]$ (middle left). The spectrum shrinks, but it still lies on both sides of the origin.

Pauli–Villars preconditioning. The operator $D_{\mathrm{PV}}^{-1}D$, with $D_{\mathrm{PV}} = D(m = 1)$, approximates the overlap operator [21]. Since D_{PV}^{-1} costs as much as a solve with D, Ref. [21] used $D_{\mathrm{PV}}^\dagger$ instead. $D_{\mathrm{PV}}^\dagger D$ has positive real parts (top right). Its low modes lie close to the imaginary axis, shifted by about the quark mass, as for an overlap operator. Its high end, however, reaches about 187, which an overlap operator would not. If we replace D_{PV} by its block-diagonal approximation B_{PV}, which drops the hops between 4^4 blocks, negative real parts come back (middle and bottom right).

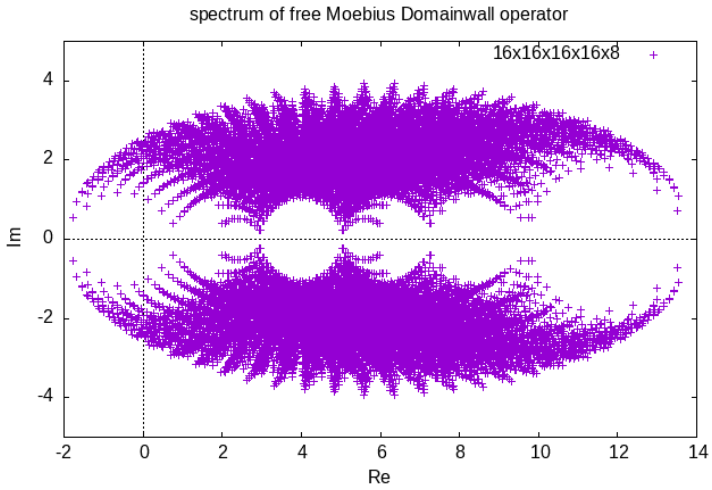


FIG. 1. Spectrum of a free Möbius domain-wall operator on $16^4 \times 8$ with quark mass 0.01, $b = 1.5$, $c = 0.5$ and domain-wall height $M_5 = 1$. Reproduced from Ref. [24], Fig. 1, under its license CC BY-NC-ND 4.0.

Both together. The operator $A' = (D_{\mathrm{PV}} C_{\mathrm{PV}}^{-1})^\dagger D C^{-1}$ has real parts in $[2.67 \times 10^{-4},\ 5.22]$ (Fig. 3, left). Its condition number, 1.96×10^4, is smaller than the 2.34×10^4 of $D_{\mathrm{PV}}^\dagger D$ and the smallest in Ref. [24]. With D_{PV} replaced by B_{PV}, the real parts become negative again (right). So a preconditioner that truncates the Pauli–Villars factor to blocks would inherit eigenvalues with negative real part. Our method keeps the factor exact on the fine lattice and approximates it only through its Galerkin coarse operator (Sec. VI).

Two caveats. First, the figures show the full-lattice operator A', not the even-odd operators of Sec. V. Second, positive real parts of the eigenvalues do not mean a positive field of values. For a non-normal operator the GCR bound [23] needs the Hermitian part $(A + A^\dagger)/2$ to be positive definite. This can fail while every eigenvalue lies in the right half-plane, as Ref. [22] discusses for $D_{\mathrm{PV}}^\dagger D$. It occurs for the even-odd form: for $\tilde{A}_R$ itself the field of values reaches just below zero on 12^4 ($\mathrm{Re}\, W \geq -4.6 \times 10^{-3}$ with stout smearing, -5.0×10^{-2} without), and for the multigrid-preconditioned $\tilde{A}_R \mathcal{M}$ far below (Sec. XII.B). The bound of Ref. [23] is then lost, but the spectrum of $\tilde{A}_R \mathcal{M}$ stays at $\mathrm{Re}\, \lambda \geq 0.78$ and its pseudospectra bound full GMRES (Secs. XII.C, XII.D); truncated GCR is not covered (Sec. XII.E). For the full-lattice A' it has not been measured.

IV. The operator and Pauli–Villars preconditioning

This section defines the Möbius operator in the form of Brower, Neff and Orginos [8] and derives from it the quantities the method is built on: the part C of the operator that acts within a four-dimensional site and its inverse, the Γ_5-transformation of D, the Pauli–Villars preconditioned operator A' of Ref. [24], and Neff's α. Every step is written out. The notation follows Ref. [8] except where stated; Appendix A relates it to Ref. [24] and to Bridge++ symbol by symbol.

A. The Möbius operator

Ingredients. ψ is a field on the four-dimensional lattice with an extra index $s = 0, \ldots, L_s - 1$, the fifth dimension; ψ_s is an ordinary Dirac field for each s. $P_\pm = \frac{1}{2}(1 \pm \gamma_5)$ are the chiral projectors. The Wilson hopping term is

$$(K\psi)(x) = \sum_{\mu=1}^{4} \Big[(1 - \gamma_\mu) U_\mu(x) \psi(x + \hat{\mu}) + (1 + \gamma_\mu) U_\mu^\dagger(x - \hat{\mu}) \psi(x - \hat{\mu}) \Big], \tag{4.1}$$

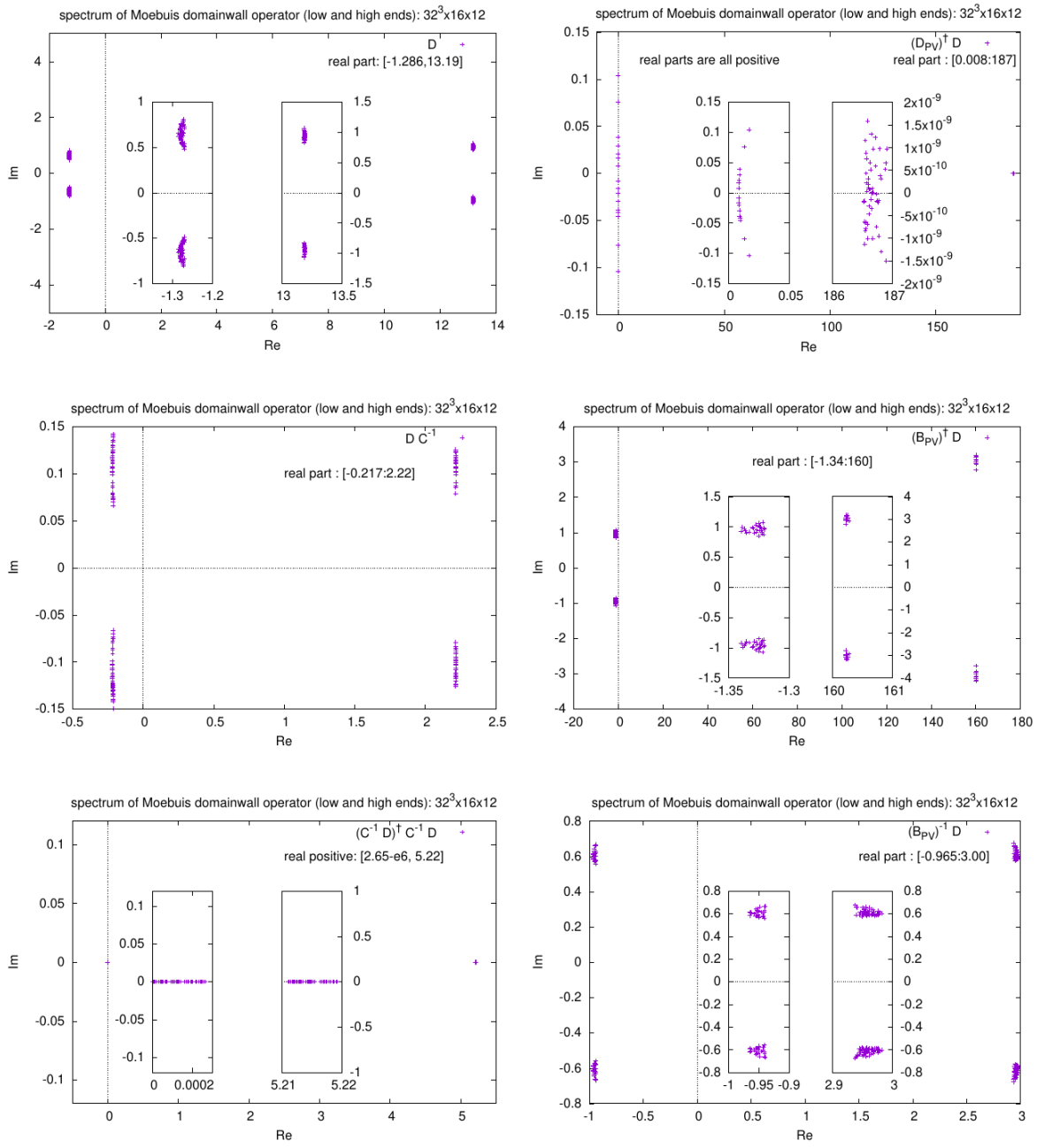


FIG. 2. Low and high ends of the spectrum of the Möbius operator on a JLQCD $2+1$-flavor configuration at the physical point, with preconditioners. Left column: D, DC^{-1}, $(C^{-1}D)^\dagger C^{-1}D$. Right column: $D_{\rm PV}^\dagger D$, $B_{\rm PV}^\dagger D$, $B_{\rm PV}^{-1}D$, where $B_{\rm PV}$ is the block-diagonal approximation of $D_{\rm PV}$ (inverted approximately in the last panel). The text of Ref. [24] quotes $48^3 \times 16$ with $L_s = 12$; the panel titles read $32^3 \times 16 \times 12$. Reproduced from Ref. [24], Fig. 3, under its license CC BY-NC-ND 4.0.

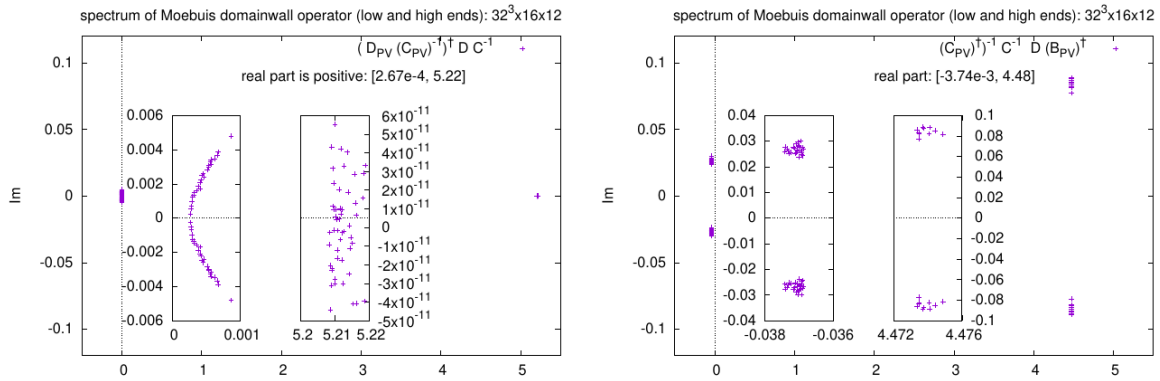


FIG. 3. Left: spectrum of $A' = (D_{\rm PV}C_{\rm PV}^{-1})^\dagger D C^{-1}$, Eq. (4.18), on the configuration of Fig. 2; all real parts are positive. Right: the same with $D_{\rm PV}$ replaced by its block-diagonal approximation $B_{\rm PV}$; negative real parts reappear. Lattice size as in Fig. 2 (the text of the source quotes $48^3 \times 16$, $L_s = 12$; the panel titles read $32^3 \times 16 \times 12$). Reproduced from Ref. [24], Fig. 4, under its license CC BY-NC-ND 4.0.

and the Wilson operator with mass $-M_5$ (the domain-wall height $M_5 > 0$) is

$$W = (4 - M_5) - \tfrac{1}{2}K. \tag{4.2}$$

This is the usual $D_W(-M_5)$ with hopping coefficient $\frac{1}{2}$; in the κ normalization $W = (4 - M_5)(1 - \kappa K)$ with $\kappa = 1/(8 - 2M_5)$. W acts on each slice s in the same way and does not mix slices. Its free spectrum, needed in Sec. XI, is $\{-M_5 + \sum_\mu (1 - \cos p_\mu) \pm i(\sum_\mu \sin^2 p_\mu)^{1/2}\}$.

The operator. With

$$D_+ = 1 + bW, \qquad D_- = cW - 1 \tag{4.3}$$

(Ref. [8], Eq. (2.18)), the Möbius operator at quark mass m acts slice by slice as

$$(D\psi)_s = D_+\psi_s + D_-\big(P_-\psi_{s+1} + P_+\psi_{s-1}\big), \qquad \psi_{L_s} \equiv -m\,\psi_0, \quad \psi_{-1} \equiv -m\,\psi_{L_s-1} \tag{4.4}$$

(Ref. [8], Eq. (2.20), with s counted from 0). For $L_s = 4$, as a matrix in s,

$$D = \begin{pmatrix} D_+ & D_-P_- & 0 & -mD_-P_+ \\ D_-P_+ & D_+ & D_-P_- & 0 \\ 0 & D_-P_+ & D_+ & D_-P_- \\ -mD_-P_- & 0 & D_-P_+ & D_+ \end{pmatrix}. \tag{4.5}$$

Shamir's operator is $b = 1$, $c = 0$; the results of this note use $b = 1.5$, $c = 0.5$ ($b - c = 1$, $b + c = 2$). The quark mass enters only through the two corners. The physical four-dimensional quark field is $q = P_-\psi_0 + P_+\psi_{L_s-1}$.

The hop in the fifth dimension. Eq. (4.4) separates into a part that hops in the fifth dimension and one that does not. Define

$$(Y\psi)_s = -P_-\psi_{s+1} - P_+\psi_{s-1} \qquad \text{(with the same wrap-around)}, \tag{4.6}$$

the off-diagonal part of Eq. (4.5) without D_- and with a minus sign. Then $D = D_+ - D_-Y$. The order matters: D_- stands to the *left* of Y, since in Eq. (4.4) D_- acts on the shifted and projected field. Inserting Eqs. (4.2) and (4.3),

$$D = 1 + bW - (cW - 1)Y = 1 + Y + W\,(b - cY) = C - \tfrac{1}{2}K\,(b - cY), \qquad C = d_+ + d_-Y, \tag{4.7}$$

with $d_+ = 1 + b(4 - M_5)$ and $d_- = 1 - c(4 - M_5)$. In the last step $W = (4 - M_5) - \frac{1}{2}K$ was split into its site-local constant and its hop; the constant parts form C. Note $d_- = -(c(4 - M_5) - 1)$ is *minus* the site-local part of D_-, because Y carries the sign. The Pauli–Villars operator is the same operator at unit mass, $D_{\rm PV} = D|_{m=1}$, $C_{\rm PV} = C|_{m=1}$; $D_\pm$ and K do not depend on m.

B. Γ_5-transformation and the exact adjoint

For the Wilson operator $\gamma_5 W \gamma_5 = W^\dagger$, so a solve with $W^\dagger$ is a solve with W between two multiplications by γ_5. The analogue for domain-wall fermions uses

$$\Gamma_5 = \gamma_5 R_5, \qquad (R_5\psi)_s = \psi_{L_s-1-s}. \tag{4.8}$$

Each building block transforms as follows.

1. W, D_+, D_-: they act on every slice alike, so they commute with R_5, and $\gamma_5 W \gamma_5 = W^\dagger$. Hence $\Gamma_5 W \Gamma_5 = W^\dagger$, $\Gamma_5 D_\pm \Gamma_5 = D_\pm^\dagger$.

2. Y: it has no hop in the four dimensions, so $\Gamma_5 Y \Gamma_5$ can be read off Eq. (4.6). γ_5 commutes with $P_\pm$; R_5 turns the hop $s \to s+1$ into $s \to s-1$. So $\Gamma_5 Y \Gamma_5$ has $-P_-$ on the hop down and $-P_+$ on the hop up, and the corner $-m$ moves to the other corner. This is the transpose of Eq. (4.6) in s with $P_\pm$ in place, which is $Y^\dagger$ since $P_\pm^\dagger = P_\pm$ and m is real: $\Gamma_5 Y \Gamma_5 = Y^\dagger$.

3. W and Y do not commute: γ_μ in K anticommutes with γ_5, so $\gamma_\mu P_\pm = P_\mp \gamma_\mu$, and the hop of K exchanges the two projectors of Y.

From $D = D_+ - D_- Y$ it follows that

$$\Gamma_5 D \Gamma_5 = D_+^\dagger - D_-^\dagger Y^\dagger, \qquad \text{whereas} \qquad D^\dagger = D_+^\dagger - Y^\dagger D_-^\dagger, \tag{4.9}$$

the adjoint reversing the order of the product. The difference is a commutator,

$$\Gamma_5 D \Gamma_5 - D^\dagger = Y^\dagger D_-^\dagger - D_-^\dagger Y^\dagger = c\,[Y^\dagger, W^\dagger] = \tfrac{c}{2}\big[K^\dagger, Y^\dagger\big], \tag{4.10}$$

where $D_-^\dagger = cW^\dagger - 1$ and $W^\dagger = (4 - M_5) - \frac{1}{2}K^\dagger$ were used. It vanishes for Shamir's operator ($c = 0$) and is of order c otherwise. Thus $\Gamma_5 D \Gamma_5 \neq D^\dagger$ for Möbius fermions: a multigrid built for D cannot be used for $D^\dagger$ by conjugating with Γ_5. On a 12^4 configuration with stout links ($M_5 = 1$, $m = 0.001$), $\|\Gamma_5 D \Gamma_5 b - D^\dagger b\| / \|D^\dagger b\| = 0.15$ for $(b, c) = (1.5, 0.5)$, 0.22 for $(2, 1)$, and 0 for $(1, 0)$, for a random b.

The exact relation. D_+ and D_- are functions of the same W and commute. Where D_- is invertible, factor it out to the left:

$$D = D_-\, K_D, \qquad K_D = D_-^{-1} D_+ - Y. \tag{4.11}$$

K_D is Γ_5-Hermitian: by item 1, $\Gamma_5 D_-^{-1} D_+ \Gamma_5 = (D_-^\dagger)^{-1} D_+^\dagger$, and this equals $(D_-^{-1} D_+)^\dagger = D_+^\dagger (D_-^\dagger)^{-1}$ because $D_+^\dagger$ and $D_-^\dagger$ commute; with item 2, $\Gamma_5 K_D \Gamma_5 = K_D^\dagger$. Then

$$D^\dagger = K_D^\dagger D_-^\dagger = \Gamma_5 K_D \Gamma_5\, D_-^\dagger = \Gamma_5 D_-^{-1} D\, \Gamma_5 D_-^\dagger, \tag{4.12}$$

and multiplying by $D_- \Gamma_5$ from the left,

$$\boxed{D_-\, \Gamma_5\, D^\dagger = D\, \Gamma_5\, D_-^\dagger} \tag{4.13}$$

This form has no inverse, holds for any m (so also for $D_{\rm PV}$, with the same D_-), and can be checked by applying both sides to a vector. On the configuration above both sides agree to 3×10^{-16} for $(b, c) = (1.5, 0.5)$ and $(2, 1)$. The Hermitian operator of the Möbius case is $\Gamma_5 K_D = \Gamma_5 D_-^{-1} D$; Ref. [8], Eq. (2.22), defines its Γ_5 as $\gamma_5 R_5 D_-^{-1}$ for that reason. It contains D_-^{-1}, whose cost is discussed in Sec. XI.

C. The site-diagonal part and its inverse

$C = d_+ + d_- Y$ of Eq. (4.7) is what remains of D when the four-dimensional hops are dropped. It acts within one four-dimensional site, as an $L_s \times L_s$ matrix in s times the identity in color, and it commutes with $P_\pm$ (only Y is involved, and Y maps $P_+\psi$ to $P_+\psi$ and $P_-\psi$ to $P_-\psi$). So it splits into a block on $\psi^+ = P_+\psi$ and one on $\psi^- = P_-\psi$. From Eq. (4.6), Y moves ψ^+ from $s-1$ to s and ψ^- from $s+1$ to s, with the factor -1, and the wrap-around gives $+m$:

$$(C\psi)^+_s = \begin{cases} d_+\psi^+_0 + m\, d_-\psi^+_{L_s-1}, & s = 0, \\ d_+\psi^+_s - d_-\psi^+_{s-1}, & s \geq 1, \end{cases} \tag{4.14}$$

a lower bidiagonal matrix with one corner element; for ψ^- the same with $s \to L_s - 1 - s$ (upper bidiagonal). For $L_s = 4$ the block on ψ^+ is

$$C^+ = \begin{pmatrix} d_+ & 0 & 0 & m\, d_- \\ -d_- & d_+ & 0 & 0 \\ 0 & -d_- & d_+ & 0 \\ 0 & 0 & -d_- & d_+ \end{pmatrix}. \tag{4.15}$$

The inverse. To solve $C^+x = r$, write the rows $s \geq 1$ as $x_s = (r_s + d_- x_{s-1})/d_+$. Starting from an unknown x_0, each x_s is affine in x_0: with $q = d_-/d_+$,

$$x_s = p_s + q^s x_0, \qquad p_0 = 0, \quad p_s = (r_s + d_- p_{s-1})/d_+, \tag{4.16}$$

which is checked by induction: $(r_s + d_-(p_{s-1} + q^{s-1}x_0))/d_+ = p_s + q^s x_0$. The row $s = 0$, $d_+ x_0 + m\, d_- x_{L_s-1} = r_0$, with $x_{L_s-1} = p_{L_s-1} + q^{L_s-1}x_0$, fixes x_0:

$$x_0 = \frac{r_0 - m\, d_-\, p_{L_s-1}}{d_+ + m\, d_-\, q^{L_s-1}}, \qquad x_s = p_s + q^s x_0. \tag{4.17}$$

For C^- the substitution runs from $s = L_s - 1$ downwards. $C^\dagger$ has the transposed blocks: on ψ^+ an upper bidiagonal matrix, solved like C^-, and on ψ^- a lower one. C^{-1} is thus exact, costs $O(L_s)$ per site, and does not depend on the gauge field [24]. It exists unless $d_+ + m\, d_- q^{L_s-1} = 0$.

D. The preconditioned operator

The spectrum of D surrounds the origin (Sec. III), so Krylov methods and polynomial smoothers cannot work on it directly. Ref. [24] multiplies $D\psi = \eta$ from the left by $(D_{\rm PV}C_{\rm PV}^{-1})^\dagger$ and substitutes $\psi = C^{-1}y$:

$$A'y = (D_{\rm PV}C_{\rm PV}^{-1})^\dagger\, \eta, \qquad A' = (D_{\rm PV}C_{\rm PV}^{-1})^\dagger\, D\, C^{-1}, \qquad \psi = C^{-1}y. \tag{4.18}$$

Why these two factors:

- DC^{-1} is D with its site-local part divided out; its eigenvalues are much more compact than those of D (Fig. 2: real parts $[-0.22, 2.22]$ against $[-1.29, 13.2]$), but still on both sides of the origin.
- For $m = 1$, $DC^{-1} = D_{\rm PV}C_{\rm PV}^{-1}$ and $A' = (D_{\rm PV}C_{\rm PV}^{-1})^\dagger(D_{\rm PV}C_{\rm PV}^{-1})$, which is Hermitian and positive definite. For $m < 1$ the two factors differ only in the corners of Eq. (4.5), and A' is a perturbation of a positive operator.
- $D_{\rm PV}^{-1}D$ approximates the overlap operator [21]; $D_{\rm PV}^\dagger$ in place of $D_{\rm PV}^{-1}$ keeps the cost of one application of A' at two applications of D plus the site-local C^{-1} and $C_{\rm PV}^{-1}$.

The result is an operator whose eigenvalues all have positive real part (Fig. 3), with the smallest condition number among those of Ref. [24].

E. Link smearing

The fermion operator may be built from smeared links in place of $U_\mu(x)$; smearing integrates the surrounding gauge field into each link and suppresses the short-distance fluctuations that produce near-zero modes of W. We use the stout projection combined with the APE staples [9]:

$$C_\mu(x) = \sum_{\nu\neq\mu} \rho\Big[U_\nu(x)U_\mu(x+\hat\nu)U_\nu^\dagger(x+\hat\mu) + U_\nu^\dagger(x-\hat\nu)U_\mu(x-\hat\nu)U_\nu(x-\hat\nu+\hat\mu)\Big], \tag{4.19}$$

$$U'_\mu(x) = \exp\Big(\big[C_\mu(x)U_\mu^\dagger(x)\big]_{\mathrm{AT}}\Big)\, U_\mu(x), \tag{4.20}$$

where $[\cdots]_{\mathrm{AT}}$ is the anti-Hermitian traceless part and ρ a tunable parameter. The step (4.20) is applied repeatedly; "stout $\rho\times n$" means n steps with the same ρ. Only W (through K) sees the smeared links; Y, C and Q do not depend on the gauge field. In HMC the force of the fermion action is taken with respect to the smeared links and carried back to U through the derivative of Eq. (4.20).

F. Neff's α

Neff's form of the domain-wall matrix [25, 26] is $D^{(\alpha)} = DQ$ with a site-local Q, diagonal in the fifth dimension,

$$Q_0 = P_- + \alpha P_+, \qquad Q_s = \alpha \;\; (0<s<L_s-1), \qquad Q_{L_s-1} = P_+ + \alpha P_-. \tag{4.21}$$

Properties used below, each read off Eq. (4.21):

- Q is Hermitian, and Q^{-1} is Eq. (4.21) with $\alpha \to 1/\alpha$.
- Q leaves the components $P_-\psi_0$ and $P_+\psi_{L_s-1}$ unchanged, which make up the quark field q; so $D^{(\alpha)}$ and D have the same four-dimensional propagator.
- The solution of $D\psi = \eta$ is $\psi = Q\psi^{(\alpha)}$ with $D^{(\alpha)}\psi^{(\alpha)} = \eta$, since $D^{(\alpha)}\psi^{(\alpha)} = DQ\psi^{(\alpha)} = D\psi$.
- Q is site-local, so the site-diagonal part of DQ is CQ, with $(CQ)^{-1} = Q^{-1}C^{-1}$ and $(CQ)^\dagger = QC^\dagger$.

The factor may be applied to the quark operator alone or to the Pauli–Villars operator as well; we write $D_{\mathrm{PV}}^{(\alpha)} = D_{\mathrm{PV}}Q$ in the latter case. For $\alpha < 1$ it improves the conditioning of the even-odd operator [25, 26].

V. Even-odd forms

The diagonal blocks are C. Order the sites so that the even ones come first. The hopping term K connects nearest neighbors, which always have opposite parity, so it has no even–even or odd–odd block. In $D = C - \frac{1}{2}K(b-cY)$ (Eq. (4.7)) the diagonal blocks therefore come from C alone and the off-diagonal ones from the hop alone:

$$D = \begin{pmatrix} D_{ee} & D_{eo} \\ D_{oe} & D_{oo} \end{pmatrix} = \begin{pmatrix} C_e & D_{eo} \\ D_{oe} & C_o \end{pmatrix}, \qquad D_{pq} = -\tfrac{1}{2}K_{pq}(b-cY) \;\; (p\neq q), \qquad C = \begin{pmatrix} C_e & 0 \\ 0 & C_o \end{pmatrix}. \tag{5.1}$$

So the D_{ee} and D_{oo} of the even-odd literature (Ref. [26], Eq. (18); Ref. [24], Eq. (4)) *are* the site-diagonal part: $D_{ee} = C_e$ and $D_{oo} = C_o$ are the blocks of the same C that appears in $A' = (D_{\mathrm{PV}}C_{\mathrm{PV}}^{-1})^\dagger DC^{-1}$, restricted to the even and to the odd sites. Since C is the same $L_s \times L_s$ matrix on every site (Sec. IV.C), C_e and C_o are also the same matrix, applied on different sites; the subscript only says where the field lives. We keep the letter C for them to make this visible, and the letter D for the off-diagonal blocks, which carry the gauge field. The same holds for D_{PV} with $C_{\mathrm{PV},e}$, $C_{\mathrm{PV},o}$.

With these blocks the Schur complement on the even sites is $S = C_e - D_{eo}C_o^{-1}D_{oe}$, and S_{PV} is that of D_{PV}. Two identities are used throughout: $S^{-1} = \pi D^{-1}\iota$ and $(S_{\mathrm{PV}}^{\dagger})^{-1} = \pi(D_{\mathrm{PV}}^{\dagger})^{-1}\iota$. To see them, write the injection and the projection of the Conventions in the same block notation:

$$\iota y = \begin{pmatrix} y \\ 0 \end{pmatrix}, \qquad \pi \begin{pmatrix} \psi_e \\ \psi_o \end{pmatrix} = \psi_e, \qquad \iota\pi \begin{pmatrix} \psi_e \\ \psi_o \end{pmatrix} = \begin{pmatrix} \psi_e \\ 0 \end{pmatrix}, \tag{5.2}$$

So $\iota\pi$ sets the odd sites to zero and leaves a field that already vanishes there unchanged. Now solve $D\psi = \iota\eta_e = (\eta_e, 0)^{\mathsf{T}}$. The odd rows give $\psi_o = -C_o^{-1}D_{oe}\psi_e$, and the even rows then read $S\psi_e = \eta_e$. Hence $S^{-1}\eta_e = \pi D^{-1}\iota\eta_e$, and the same argument for $D_{\mathrm{PV}}^{\dagger}$ gives the second identity.

Two forms. The site-diagonal factor can stand on either side of S. This gives the right and the left form,

$$\tilde{A}_R = (S_{\mathrm{PV}}C_{\mathrm{PV},e}^{-1})^{\dagger}\,(S\,C_e^{-1}), \qquad \tilde{A}_L = (C_{\mathrm{PV},e}^{-1}S_{\mathrm{PV}})^{\dagger}\,(C_e^{-1}S). \tag{5.3}$$

$\tilde{A}_R$ is the even-odd counterpart of A'. $C_e^{-1}S$ is the usual left-preconditioned even-odd operator, which Boyle and Yamaguchi found worse than S itself for CG on the normal equation [20].

With Neff's α. Replace D by DQ. Since Q acts within a site and is diagonal in s, the blocks of DQ are C_eQ_e, $D_{eo}Q_o$, $D_{oe}Q_e$ and C_oQ_o. Its Schur complement is therefore

$$C_eQ_e - D_{eo}Q_o(C_oQ_o)^{-1}D_{oe}Q_e = SQ_e, \tag{5.4}$$

and its diagonal block is C_eQ_e. In the right form the two factors of Q_e cancel: $SQ_e(C_eQ_e)^{-1} = SC_e^{-1}$. So $\tilde{A}_R$ does not depend on α. In the left form they do not cancel. With $(C_eQ_e)^{-1}SQ_e = Q_e^{-1}C_e^{-1}SQ_e$ and Q Hermitian,

$$\tilde{A}_L^{(\alpha)} = \begin{cases} Q_e\,(C_{\mathrm{PV},e}^{-1}S_{\mathrm{PV}})^{\dagger}\,Q_e^{-2}\,(C_e^{-1}S)\,Q_e, & \alpha \text{ in } D \text{ and } D_{\mathrm{PV}}, \\ (C_{\mathrm{PV},e}^{-1}S_{\mathrm{PV}})^{\dagger}\,Q_e^{-1}\,(C_e^{-1}S)\,Q_e, & \alpha \text{ in } D \text{ only}. \end{cases} \tag{5.5}$$

For $D\psi = \eta$, with $\eta'_e = \eta_e - D_{eo}C_o^{-1}\eta_o$, the right-hand side is $\rho_R = (S_{\mathrm{PV}}C_{\mathrm{PV},e}^{-1})^{\dagger}\eta'_e$ or $\rho_L = (C_{\mathrm{PV},e}^{-1}S_{\mathrm{PV}})^{\dagger}C_e^{-1}\eta'_e$ (operators with Q when $\alpha \neq 1$); the solution is $\psi_e^{(\alpha)} = C_e^{-1}y$ (right) or y (left), $\psi_o^{(\alpha)} = C_o^{-1}(\eta_o - D_{oe}\psi_e^{(\alpha)})$, $\psi = Q\psi^{(\alpha)}$.

Why both forms. The two forms serve different purposes. The factorized correction of Sec. VI is exact for the right form (Proposition 1), but Q cancels in it, so the right form cannot use Neff's α. The left form keeps α. The correction is not exact for it, yet it converges within a few iterations of the right form, and with α it can save iterations. The outer solver therefore uses the left form in practice. The measurements of Sec. XII are on the left form unless stated.

VI. The factorized coarse-grid correction and the prolongation

A. Why factorize

Solving on the even sites halves the problem and improves the conditioning. But the Schur complement reaches two hops, so the existing methods make a choice. HDCG coarsens the squared Schur complement and accepts a larger coarse stencil [18]. The methods with a small coarse stencil coarsen an operator on the whole lattice instead: $\gamma_5 R_5 D$ (HDCR [19]) or $D_{\mathrm{PV}}^{\dagger}D$ [21, 22]. The factorized correction takes both advantages. The outer operator is the even-odd one, while the coarse operators are those of D and D_{PV} themselves (Table 1).

The point counts follow from the hops. An operator that reaches k hops on the fine lattice couples coarse sites whose coordinates differ by at most 1 in at most k directions. For $k = 1$ and 2 this gives $1 + 8 = 9$ and $1 + 8 + 24 = 33$ points, and for $k \geq 4$ all $3^4 = 81$.

TABLE 1. Multigrid methods for domain-wall fermions: the system solved, the operator that is coarsened, and its coarse stencil, for blocks that contain the whole fifth dimension and are at least two sites wide in each direction (4^4 for HDCG). S is the Schur complement on the even sites.

method	system solved	operator coarsened	coarse stencil
HDCG [18, 16]	even sites, $S^\dagger S$	$S^\dagger S$ (four hops)	81 points
HDCR [19]	whole lattice, $\gamma_5 R_5 D$	$\gamma_5 R_5 D$, $c = 0$ (one hop)	9 points
Boyle and Oare [22]	whole lattice, $D_{\mathrm{PV}}^\dagger D$	$D_{\mathrm{PV}}^\dagger D$ (two hops)	33 points
this note	even sites, $\tilde{A}_R$ or $\tilde{A}_L$	D and D_{PV} apart (one hop each)	9 points each

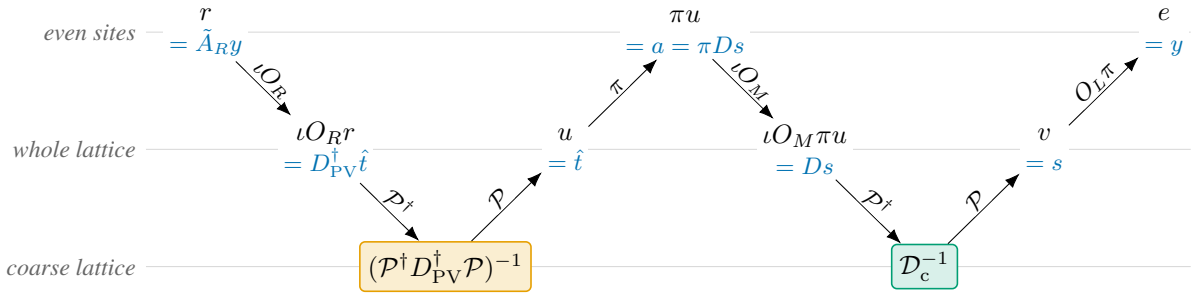


FIG. 4. One factorized coarse correction, Eq. (6.2) read from the right, on its three lattices (the Pauli–Villars coarse solve in orange, the quark one in green): $u = \mathcal{P}(\mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger \mathcal{P})^{-1} \mathcal{P}^\dagger \iota O_R r$ is the result of the first coarse solve, that of the adjoint Pauli–Villars operator, and its even part, with O_M, is the right-hand side of the second, that of the quark operator, whose result is $v = \mathcal{P}\mathcal{D}_{\mathrm{c}}^{-1}\mathcal{P}^\dagger \iota O_M \pi u$; $e = O_L \pi v$. In blue, after "=", the values for $r = \tilde{A}_R y$ in the right form with exact coarse solves and $s, \hat{t} \in \operatorname{range} \mathcal{P}$ (Proposition 1; $O_R = C_{\mathrm{PV},e}^\dagger$, $O_M = 1$, $O_L = C_e$): the first solve returns the lift $\hat{t}$, whose even part is $a = \pi D s$, the second returns the lift s, and the correction returns y.

The correction. By $S^{-1} = \pi D^{-1} \iota$, the inverse of the right form is

$$\tilde{A}_R^{-1} = C_e \, S^{-1} \, (S_{\mathrm{PV}}^\dagger)^{-1} \, C_{\mathrm{PV},e}^\dagger = C_e \, \pi D^{-1} \iota \; \pi (D_{\mathrm{PV}}^\dagger)^{-1} \iota \; C_{\mathrm{PV},e}^\dagger, \tag{6.1}$$

a product of two inverses of one-hop operators and site-local factors. The Galerkin operator of $\tilde{A}_R$ itself would have a large coarse stencil, since $\tilde{A}_R$ reaches four hops. The Galerkin operators of D and D_{PV} are nearest-neighbor, since D is nearest-neighbor in four dimensions and every aggregate contains the whole fifth dimension. We therefore replace each inverse by its Galerkin approximation $\mathcal{P}(\mathcal{P}^\dagger X \mathcal{P})^{-1}\mathcal{P}^\dagger$. For both forms and any α, the correction of a residual r on the even sites is

$$e = O_L \; \pi\mathcal{P} \, \mathcal{D}_{\mathrm{c}}^{-1} \mathcal{P}^\dagger \iota \;\; O_M \;\; \pi\mathcal{P} \left(\mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger \mathcal{P}\right)^{-1} \mathcal{P}^\dagger \iota \;\; O_R \, r, \qquad \mathcal{D}_{\mathrm{c}} = \mathcal{P}^\dagger D \mathcal{P}, \tag{6.2}$$

with the site-local factors O_R, O_M, O_L of Table 2. They come from writing each $\tilde{A}^{-1}$ as in Eq. (6.1). For example, for the left form with α in both operators, Eq. (5.5) gives

$$\tilde{A}_L^{(\alpha)\,-1} = Q_e^{-1} \, S^{-1} \, C_e Q_e^2 C_{\mathrm{PV},e}^\dagger \, (S_{\mathrm{PV}}^\dagger)^{-1} \, Q_e^{-1}, \tag{6.3}$$

so $O_R = Q_e^{-1}$, $O_M = C_e Q_e^2 C_{\mathrm{PV},e}^\dagger$ and $O_L = Q_e^{-1}$. The coarse operators are always those of the plain D and D_{PV}, without α, so one setup serves both forms and every α. Figure 4 follows a residual through Eq. (6.2). It moves between the even sites, the whole lattice and the coarse lattice twice, and the result of the first coarse solve, taken on the even sites, is the right-hand side of the second.

B. Aggregation and prolongation

We divide the fine lattice into rectangular four-dimensional blocks. Each block contains the whole fifth dimension and becomes one coarse site. We split each test vector v into its Γ_5 eigencomponents

TABLE 2. Site-local factors of the correction (6.2). They follow from $\tilde{A}^{-1}$ with $S^{-1} = \pi D^{-1}\iota$; with α the site-diagonal blocks are $C_e Q_e$ and $Q_e C_{\mathrm{PV},e}^\dagger$.

form	O_R	O_M	O_L
right, any α	$C_{\mathrm{PV},e}^\dagger$	1	C_e
left, α in D only	1	$C_e Q_e C_{\mathrm{PV},e}^\dagger$	Q_e^{-1}
left, α in D and D_{PV}	Q_e^{-1}	$C_e Q_e^2 C_{\mathrm{PV},e}^\dagger$	Q_e^{-1}

$v_\pm = \frac{1}{2}(1 \pm \Gamma_5)v$ and orthonormalize the $+$ parts and the $-$ parts separately within each block. With n_v test vectors a coarse site then carries $2n_v$ components: the first n_v of positive and the last n_v of negative Γ_5-chirality.

This split makes $\mathcal{P}$ compatible with Γ_5. Each column of $\mathcal{P}$ is an eigenvector of Γ_5 with eigenvalue $+1$ or -1, so with $\Sigma = \mathrm{diag}(\mathbf{1}_{n_v}, -\mathbf{1}_{n_v})$ on each coarse site,

$$\Gamma_5 \mathcal{P} = \mathcal{P}\Sigma. \tag{6.4}$$

It follows that $\mathcal{P}^\dagger(\Gamma_5 X \Gamma_5)\mathcal{P} = \Sigma\, \mathcal{P}^\dagger X \mathcal{P}\, \Sigma$ for any fine operator X. In particular $\mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger \mathcal{P} = \Sigma \mathcal{D}_{\mathrm{c}}^{\mathrm{PV}} \Sigma$ with $\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}} = \mathcal{P}^\dagger D_{\mathrm{PV}} \mathcal{P}$ whenever $\Gamma_5 D_{\mathrm{PV}} \Gamma_5 = D_{\mathrm{PV}}^\dagger$, that is for $c = 0$.

The split also halves the storage. Since $\Gamma_5 = \gamma_5 R$, the condition $\Gamma_5 v_\pm = \pm v_\pm$ reads $v_\pm(L_s - 1 - s) = \pm\gamma_5 v_\pm(s)$. So each part is fixed by its values for $s < L_s/2$, and $\mathcal{P}$ is stored only there. On the even sites the transfer operators are $\mathcal{P}^\dagger\iota$ and $\pi\mathcal{P}$.

C. The basis $\{s, \hat{t}\}$

The prolongation is built from vectors y_j on the even sites that approximate low modes of $\tilde{A}_R$ (Sec. VIII). Two facts shape how we use them.

- $\mathcal{P}$ and the coarse operators act on the whole lattice, so each y must be extended, or *lifted*, to the whole lattice. We fill the odd sites so that the fine operator maps the lifted vector to the even-odd quantity, just as ψ_o is rebuilt from ψ_e after an even-odd solve.
- The correction (6.2) contains two Galerkin solves, one with $D_{\mathrm{PV}}^\dagger$ and one with D. Each is exact only on vectors in $\mathrm{range}\,\mathcal{P}$. So we lift each y in two ways, one for each solve.

The two lifts are

$$s = \big(C_e^{-1} y,\ -C_o^{-1} D_{oe} C_e^{-1} y\big), \tag{6.5}$$

$$\hat{t} = \big(a,\ -(C_{\mathrm{PV},o}^\dagger)^{-1}(D_{\mathrm{PV},eo})^\dagger a\big), \qquad a = \pi D s. \tag{6.6}$$

The lift s is chosen so that Ds vanishes on the odd sites. With the blocks of Eq. (5.1),

$$Ds = \begin{pmatrix} C_e C_e^{-1} y - D_{eo} C_o^{-1} D_{oe} C_e^{-1} y \\ D_{oe} C_e^{-1} y - C_o C_o^{-1} D_{oe} C_e^{-1} y \end{pmatrix} = \begin{pmatrix} S C_e^{-1} y \\ 0 \end{pmatrix} = \iota\, S C_e^{-1} y. \tag{6.7}$$

In the same way $\hat{t}$ is chosen so that $D_{\mathrm{PV}}^\dagger \hat{t}$ vanishes on the odd sites. The block form of $D_{\mathrm{PV}}^\dagger$ has $(D_{\mathrm{PV},eo})^\dagger$ as its odd-even block and $C_{\mathrm{PV},o}^\dagger$ as its odd diagonal block, and the same computation gives $D_{\mathrm{PV}}^\dagger \hat{t} = \iota\, S_{\mathrm{PV}}^\dagger a$.

The first coarse solve of the correction returns $\hat{t}$ and the second returns s (Proposition 1 below). Since $\hat{t}$ is made from Ds, the two lifts follow the order of the solves. With the y alone in the basis, each factor of the correction would be an approximation, and the errors of the two would multiply. The $2n$ vectors $\{s_j, \hat{t}_j\}$ are the test vectors, $n_v = 2n$.

The structure is familiar from block preconditioners for the incompressible Navier–Stokes equations [33]. There the velocity is eliminated, as the odd sites are here, and the pressure Schur complement is approximated by a product of simpler operators, each solved by its own multigrid. In the least-squares commutator preconditioner, $(BB^{\mathsf{T}})^{-1}(BFB^{\mathsf{T}})(BB^{\mathsf{T}})^{-1}$, these are two Poisson-like inverses. The product is only as good as each factor on the vectors it is applied to; here those are $\hat{t}$ and s.

Proposition 1 (Exactness, right form)**.** *Let $s, \hat{t} \in \operatorname{range}\mathcal{P}$, let $\mathcal{D}_{\mathrm{c}}$ and $\mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger \mathcal{P}$ be invertible, and let both coarse systems in Eq. (6.2) be solved exactly, with the exact adjoint $\mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger \mathcal{P} = (\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}})^\dagger$. Then the right-form correction of $r = \tilde{A}_R y$ is $e = y$.*

Proof. For any invertible $\mathcal{P}^\dagger X \mathcal{P}$, the map $\mathcal{P}(\mathcal{P}^\dagger X \mathcal{P})^{-1}\mathcal{P}^\dagger X$ is the identity on $\operatorname{range}\mathcal{P}$. We follow r through the correction with the factors of the right form, $O_R = C_{\mathrm{PV},e}^\dagger$, $O_M = 1$, $O_L = C_e$.

1. $O_R r = C_{\mathrm{PV},e}^\dagger \tilde{A}_R y = S_{\mathrm{PV}}^\dagger S C_e^{-1} y = S_{\mathrm{PV}}^\dagger a$, and $\iota S_{\mathrm{PV}}^\dagger a = D_{\mathrm{PV}}^\dagger \hat{t}$. So the first coarse solve returns $\hat{t}$, whose even part is $\pi\hat{t} = a = \pi D s$.

2. The second solve has the right-hand side $\mathcal{P}^\dagger \iota \pi D s = \mathcal{P}^\dagger D s$, because Ds vanishes on the odd sites. It returns s, whose even part is $\pi s = C_e^{-1} y$.

3. Finally $O_L = C_e$ gives $e = y$. □

Remark 1 (Conditions in practice). The proposition needs three things. (i) s and $\hat{t}$ in the range of the stored $\mathcal{P}$. This holds only up to the rounding of the reduced-precision storage of $\mathcal{P}$. (ii) The exact adjoint of the Pauli–Villars coarse factor. The Σ-adjoint $\Sigma\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}}\Sigma$ of Sec. VII equals it only if $\Gamma_5 D_{\mathrm{PV}} \Gamma_5 = D_{\mathrm{PV}}^\dagger$, which fails for $c \neq 0$ by Eq. (4.10). (iii) Exact coarse solves, whereas we stop them at a loose tolerance. Numerically, with both coarse factors solved on their exact adjoints to $|r|^2/|b|^2 < 10^{-24}$ in double precision and $\mathcal{P}$ in half precision, the correction of $r = \tilde{A}_R y_j$ returns y_j to $\|e - y_j\|/\|y_j\| = 3 \times 10^{-4}$–$5 \times 10^{-4}$ on $4^3 \times 8$ (2^4 aggregates, $M_5 = 1.8$, $m = 0.05$) and 3×10^{-4}–2.5×10^{-3} on 12^4 (setup S, $m = 0.001$). For random vectors the same quantity is 3–37. So the proposition holds to the rounding of $\mathcal{P}$, and (i) costs about 10^{-3}. (ii) costs nothing measurable (Sec. XII.G). (iii) even helps: with GCR(2), accurate coarse solves need more iterations or stagnate (Secs. XII.E, XII.H).

Remark 2 (Left form). For the left form with α in both operators, the same argument needs the quark lift s' of $Q_e y'$ and the Pauli–Villars lift $\hat{t}'$ of $u = C_{\mathrm{PV},e}^{\dagger\,-1} Q_e^{-2} C_e^{-1} S Q_e y'$ in $\operatorname{range}\mathcal{P}$ (Table 2 gives $O_M u = S Q_e y'$). For $\alpha = 1$ and $y' = C_e^{-1} y$ we have $s' = s$. But $u = C_{\mathrm{PV},e}^{\dagger\,-1} C_e^{-1} a$ differs from a by a site-local matrix that mixes s and spin, so $\hat{t}'$ is in general not in the span of the right-form vectors. The correction is therefore not exact for the left form. Numerically ($\alpha = 1$), for $r = \tilde{A}_L y'$ with $y' = C_e^{-1} y_j$ the correction misses y' by $\|e - y'\|/\|y'\| = 0.05$–0.18 on $4^3 \times 8$ and 1.6–24 on 12^4, not much better than for random vectors (8–35). The left form converges nonetheless: it has the spectrum of the right form and needs one to five iterations more or fewer (Secs. XII.C, XII.E, XII.H). So the cycle does not need the exactness of Proposition 1.

D. The coarse operators

We assemble $\mathcal{D}_{\mathrm{c}} = \mathcal{P}^\dagger D \mathcal{P}$ and $\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}} = \mathcal{P}^\dagger D_{\mathrm{PV}} \mathcal{P}$ from the hops of the fine operator between neighboring blocks and from its part within a block. Each coarse site has a $2n_v \times 2n_v$ diagonal block and one link block per direction, nine blocks in all. We do not assemble the adjoint factor $\mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger \mathcal{P}$. We apply it either exactly, as the conjugate transpose $(\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}})^\dagger$ of the stored blocks, or approximately as $\Sigma\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}}\Sigma$. When all extents of the coarse lattice are even, the coarse hops connect coarse sites of opposite parity, and both coarse operators have an even-odd form of their own.

VII. The coarse solvers

We solve both coarse systems only approximately, in reduced precision, to a loose relative residual or a maximum number of iterations. Reaching the tolerance is not required. The two are solved in sequence: first the Pauli–Villars factor, whose solution enters the right-hand side of the quark factor. The default, used for all results unless stated, is BiCGStab on the even-odd form of the quark factor and MINRES on the Σ-Hermitian form of the Pauli–Villars factor, both to $|r|^2/|b|^2 < 10^{-2}$. This section describes these and the alternatives.

A. The quark factor

Write $\mathcal{D}_{\mathrm{c}} = \begin{pmatrix} A_{ee} & B_{eo} \\ B_{oe} & A_{oo} \end{pmatrix}$ on the coarse lattice. The system $\mathcal{D}_{\mathrm{c}} c = g$ is solved through its Schur complement,

$$\left(A_{ee} - B_{eo} A_{oo}^{-1} B_{oe}\right) c_e = g_e - B_{eo} A_{oo}^{-1} g_o, \qquad c_o = A_{oo}^{-1}(g_o - B_{oe} c_e), \tag{7.1}$$

with BiCGStab or truncated flexible GCR, preconditioned by the exact inverses of the diagonal blocks A_{ee}^{-1}. We need a non-Hermitian Krylov method here: $\mathcal{D}_{\mathrm{c}}$ is not Hermitian, and $\Sigma\mathcal{D}_{\mathrm{c}}$ is Hermitian only for $c = 0$.

B. The Pauli–Villars factor

The system is $(\mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger \mathcal{P})\, w = g$. Two treatments are available.

Exact adjoint. Solve $(\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}})^\dagger w = g$ with a non-Hermitian Krylov method, preconditioned by the inverses of the adjoint diagonal blocks, $(A_{xx}^\dagger)^{-1} = (A_{xx}^{-1})^\dagger$.

Σ-Hermitian form. Replace the adjoint by $\Sigma\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}}\Sigma$ and solve the equivalent system

$$\mathcal{H}_{\mathrm{c}}\, w = \Sigma g, \qquad \mathcal{H}_{\mathrm{c}} = \mathcal{D}_{\mathrm{c}}^{\mathrm{PV}}\Sigma, \tag{7.2}$$

with MINRES. $\mathcal{H}_{\mathrm{c}}$ is Hermitian (indefinite) exactly when $(\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}})^\dagger = \Sigma\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}}\Sigma$, that is for $c = 0$ by Eqs. (4.10) and (6.4). For $c \neq 0$ MINRES works on a non-Hermitian matrix and has no guarantee. On the coarse space the non-Hermitian part is a few percent, up to 14% in the directions of the smallest singular values, and linear in c; MINRES on $\mathcal{H}_{\mathrm{c}}$ nevertheless needs the same outer iterations at $c = 0.5$ as for the Shamir operator and is better than the exact adjoint with BiCGStab (Sec. XII.G). The absolute-value block preconditioner and the additive level preconditioner below, also exact only for $c = 0$, are used as they are; their losses were not measured separately.The preconditioner must be Hermitian positive definite; we use the absolute value of the diagonal blocks [27], $|\mathcal{H}_{xx}|^{-1} = V|\Lambda|^{-1}V^\dagger$ with $\frac{1}{2}(\mathcal{H}_{xx} + \mathcal{H}_{xx}^\dagger) = V\Lambda V^\dagger$ on each coarse site. On a coarse lattice with even extents, the Schur complement of $\mathcal{H}_{\mathrm{c}}$ on the even sites is $S_{\mathrm{c}}\Sigma_e$, with S_{c} the Schur complement of $\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}}$; one solves for $u_e = \Sigma w_e$ on the even sites, completes u on the odd sites as in Eq. (7.1), and sets $w = \Sigma u$.

C. Levels below the coarse lattice

Both factors may be preconditioned by further levels. The coarse sites are grouped into super-blocks; on each super-block, local vectors are formed from the coarse test vectors $\mathcal{P}^\dagger s_j$ and $\mathcal{P}^\dagger \hat{t}_j$, split by Σ and orthonormalized (Gram–Schmidt applied twice); the level operators are obtained by probing the nearest-neighbor stencil; the last level is inverted densely. With Q_ℓ the prolongation into level $\ell - 1$ and B_ℓ the diagonal blocks of level ℓ:

Quark factor. A V-cycle with the coarse correction followed by one block-Jacobi sweep on the updated residual, without pre-smoothing,

$$z = Q_{\ell+1} M_{\ell+1} Q_{\ell+1}^\dagger r, \qquad M_\ell r = z + B_\ell^{-1}(r - \mathcal{D}_\ell z); \tag{7.3}$$

on the coarse lattice itself the same multiplicative combination is used, and with the even-odd coarse form its even part preconditions the Schur system. The level test vectors can optionally be refined by a few GCR steps preconditioned by the levels already built.

Pauli–Villars factor (Σ-Hermitian form only). The additive Hermitian positive preconditioner

$$M_\ell = |B_\ell|^{-1} + Q_{\ell+1} M_{\ell+1} Q_{\ell+1}^\dagger, \qquad M_{\ell_{\max}} = |\mathcal{H}_{\ell_{\max}}|^{-1}, \tag{7.4}$$

in the spirit of absolute-value preconditioning [27].

VIII. Setup

The setup produces the vectors $y_1, \dots, y_n$ on the even sites, then the basis $\{s_j, \hat{t}_j\}$, $\mathcal{P}$ and the coarse operators. It always uses the right form; the same setup serves the left forms and every α (Sec. VI). It has two stages. A Chebyshev filter finds approximate low modes without any multigrid. A bootstrap then improves them with the cycle built from them, so that they also cover what the first coarse space misses.

A. Chebyshev-filtered subspace iteration

The vectors approximate the right singular vectors of the smallest singular values of $\tilde{A}_R$, that is the low eigenvectors of the Hermitian positive operator $H_n = \tilde{A}_R^\dagger \tilde{A}_R$. We find them by filtering. Let T_d be the Chebyshev polynomial of degree d and $\ell(x) = \big(2x - (\lambda_{\rm hi} + \lambda_{\rm lo})\big)/(\lambda_{\rm hi} - \lambda_{\rm lo})$. Then $|T_d(\ell(x))| \le 1$ on $[\lambda_{\rm lo}, \lambda_{\rm hi}]$, while it grows quickly below $\lambda_{\rm lo}$. So $T_d(\ell(H_n))$ amplifies the low modes against the rest. We apply it by the three-term recurrence.

Algorithm 1. LOWMODES: Chebyshev-filtered subspace iteration on $H_n = \tilde{A}_R^\dagger \tilde{A}_R$.

1: $X \leftarrow n$ random vectors on the even sites

2: $\lambda_{\rm hi} \leftarrow$ a bound slightly above $\lambda_{\max}(H_n)$ (power iteration); $\lambda_{\rm lo} \leftarrow$ a small fraction of $\lambda_{\rm hi}$

3: **for** each round **do**

4: $d \leftarrow$ a degree proportional to $\sqrt{\lambda_{\rm hi}/\lambda_{\rm lo}}$, within fixed bounds

5: $X \leftarrow T_d(\ell(H_n))\, X$; orthonormalize X through its Gram matrix

6: $G \leftarrow X^\dagger H_n X = V \Theta V^\dagger$; $X \leftarrow XV$ ▷ Rayleigh–Ritz

7: $\lambda_{\rm lo} \leftarrow \theta_{\max}$ ▷ largest Ritz value

8: **end for**

9: **return** X, Θ

B. Bootstrap

Each bootstrap round has three steps. It builds the basis and the two-grid cycle of Sec. IX from the current y. It replaces every y_j by a few FGMRES steps on $\tilde{A}_R x = y_j$ from $x = 0$, preconditioned by that cycle. And it orthonormalizes the new y through their Gram matrix ($y \leftarrow yV\Lambda^{-1/2}$ with $y^\dagger y = V\Lambda V^\dagger$). The cycle damps the components it already handles, so the new vectors are enriched in what it does not.

Algorithm 2. Setup of the two-grid method.

1: $y \leftarrow \textsc{LowModes}(H_n)$
2: **for** each bootstrap round **do**
3: build $\mathcal{P}$ from $\{s_j, \hat{t}_j\}$ of y and the cycle $\mathcal{M}$ on it
4: **for** $j = 1, \dots, n$ **do**
5: $y_j \leftarrow$ a few FGMRES steps on $\tilde{A}_R x = y_j$, preconditioned by $\mathcal{M}$
6: **end for**
7: orthonormalize y
8: **end for**
9: compute $s_j, \hat{t}_j$ by Eqs. (6.5)–(6.6); split by Γ_5; orthonormalize per block $\to \mathcal{P}$
10: assemble $\mathcal{D}_c$, $\mathcal{D}_c^{\mathrm{PV}}$, the inverses of their diagonal blocks, their even-odd forms
11: optionally, build the levels below from $\mathcal{P}^\dagger s_j$, $\mathcal{P}^\dagger \hat{t}_j$

IX. The Chebyshev smoother and the cycle

A. Smoother

The smoother is ν steps of Chebyshev iteration [28, 29] for $\tilde{A}x = b$ from $x = 0$, i.e. a fixed polynomial $x = p(\tilde{A})b$ of degree $\nu - 1$. It is defined by an ellipse with real foci $[\lambda_{\mathrm{lo}}, \lambda_{\mathrm{hi}}]$: λ_{hi} slightly above an estimate of the spectral radius of $\tilde{A}$ from a power iteration, and $\lambda_{\mathrm{lo}} = \lambda_{\mathrm{hi}}/\kappa_s$ for a chosen ratio κ_s. With $\theta = \frac{1}{2}(\lambda_{\mathrm{hi}} + \lambda_{\mathrm{lo}})$, $\delta = \frac{1}{2}(\lambda_{\mathrm{hi}} - \lambda_{\mathrm{lo}})$, $\sigma = \theta/\delta$, $\varrho_0 = 1/\sigma$, $r_0 = b$ and $d_0 = r_0/\theta$, step k is

$$x \leftarrow x + d_k, \quad r_{k+1} = r_k - \tilde{A}d_k, \quad \varrho_{k+1} = \frac{1}{2\sigma - \varrho_k}, \quad d_{k+1} = \varrho_{k+1}\varrho_k\, d_k + \frac{2\varrho_{k+1}}{\delta}\, r_{k+1}, \tag{9.1}$$

The last step only updates x, so ν steps cost $\nu - 1$ applications of $\tilde{A}$. We use only a real focal interval. The ratio κ_s is not derived from the operator; it has to be chosen. The polynomial damps the part of the spectrum above λ_{lo} and leaves the rest to the coarse correction, and an interval that is too wide or too narrow can make the outer solve stagnate. The two setups of Sec. XII need $\kappa_s \approx 10$ ($M_5 = 1$, stout) and $\kappa_s \approx 30$ ($M_5 = 1.8$); the first stagnates with the second's interval, the second needs nearly three times the iterations with the first's (Sec. XII.H).

B. Cycle

Algorithm 3. Two-grid cycle $y = \mathcal{M}b$ for $\tilde{A}$.

1: $y \leftarrow p(\tilde{A})\, b$ ▷ pre-smoothing
2: $r \leftarrow b - \tilde{A}y$
3: $y \leftarrow y + e(r)$ ▷ correction (6.2)
4: $r \leftarrow b - \tilde{A}y$
5: $y \leftarrow y + p(\tilde{A})\, r$ ▷ post-smoothing

One cycle costs 2ν applications of $\tilde{A}$ (two smoothings and two residuals) plus the two coarse solves. Because the coarse solves are Krylov iterations stopped by a residual test, $\mathcal{M}$ is not a fixed linear map.

X. The outer solver

The outer iteration solves $\tilde{A}y = \rho$ with the cycle as a right preconditioner, and we rebuild ψ as in Sec. V. The cycle $\mathcal{M}$ changes from one application to the next, so the outer method must be flexible: FGMRES or flexible GCR, which keep the preconditioned directions. BiCGStab with the cycle is also available, but its convergence theory has no flexible counterpart.

Flexibility alone gives no convergence guarantee. For a fixed $\mathcal{M}$, restarted or truncated methods converge for every restart length if the field of values of $\tilde{A}\mathcal{M}$ lies in the right half-plane [23]. Otherwise only the full method is covered, by the spectrum and the pseudospectra of $\tilde{A}\mathcal{M}$ [15, 31], and a truncated method can stagnate [32]. For the present cycle the field of values is not in the right half-plane, the pseudospectra do bound the full method, and GCR(2) needs the coarse solves inexact: with accurate ones it needs more iterations or stagnates (Secs. XII.B–XII.E). Block variants treat several right-hand sides together. We monitor convergence on the even-odd system and evaluate the residual of the original equation, $\|D\psi - \eta\|/\|\eta\|$, with the plain D after reconstruction. The reference solver is CG on the normal equation of the even-odd operator $C_e^{-1}S$ (CGNE). It is robust but squares the condition number. It runs in double precision, or in mixed precision: CG in single precision with reliable updates in double precision. The solves with $D^\dagger$ that HMC needs use the same cycle between two Γ_5 (Sec. XI).

XI. Solving with $D^\dagger$

HMC needs $(D^\dagger D)^{-1}$ at every force evaluation, so it solves with $D^\dagger$ as often as with D. This section shows that the multigrid of D also solves with $D^\dagger$. No second setup, coarse operator or cycle is needed. The tool is the exact relation (4.13) between D and $D^\dagger$.

A. The idea

For Wilson fermions $D_W^\dagger = \gamma_5 D_W \gamma_5$, so $D_W^{-\dagger} = \gamma_5 D_W^{-1} \gamma_5$, and a solver for D_W solves with $D_W^\dagger$ as well. For Möbius fermions this fails at order c (Sec. IV.B). We use $\Gamma_5 D^{-1} \Gamma_5$ anyway, not as the inverse of $D^\dagger$ but as a preconditioner for it. The question is how good a preconditioner it is.

Multiply $D^\dagger$ by $\Gamma_5 D^{-1} \Gamma_5$ and use $D^\dagger = \Gamma_5 D_-^{-1} D \, \Gamma_5 D_-^\dagger$ from Eq. (4.13) and $\Gamma_5 D_-^\dagger \Gamma_5 = D_-$:

$$D^\dagger \, \Gamma_5 D^{-1} \Gamma_5 = \Gamma_5 D_-^{-1} D \, D_- D^{-1} \, \Gamma_5 = \Gamma_5 \big(1 + D_-^{-1} [D, D_-] D^{-1}\big) \Gamma_5. \tag{11.1}$$

The commutator follows from $D = D_+ - D_- Y$ and $[D_+, D_-] = 0$: $[D, D_-] = -D_- [Y, D_-] = -c \, D_- [Y, W]$. The factor D_- cancels, and

$$\boxed{D^\dagger \, \Gamma_5 D^{-1} \Gamma_5 = \Gamma_5 \, (1 - N) \, \Gamma_5, \qquad N = c \, [Y, W] \, D^{-1}.} \tag{11.2}$$

Three remarks.

- D_-^{-1} has dropped out. We never invert D_-, which we cannot do (Sec. XI.D).
- $[Y, W] = -\frac{1}{2}[Y, K]$ is a bounded operator: one hop in four dimensions and one in the fifth. So N is large only where D^{-1} is large, on the low modes of D.
- On a low mode v the commutator acts through the γ_μ in K, which flip chirality. A low mode of D lives on one wall with one chirality; $[Y, W]v$ lives on the other wall with the other chirality, far from the low modes. So N takes the low modes out of their span, and N^2 is small. Then $(1 - N)^{-1} \approx 1 + N$, and a Krylov method needs only a few steps more than for the identity.

The third remark is a heuristic; the measurements below test it.

B. On the even sites

The solves of HMC are on the even sites, so we need the same idea for the Schur complement S. $D^\dagger$ has the Schur complement $S^\dagger$ (Sec. XI.C), and $S^{-1} = \pi D^{-1} \iota$. Because Γ_5 acts within a site, it commutes with π and ι, and

$$S^{-\dagger} = \pi D^{-\dagger} \iota \approx \pi\, \Gamma_5 D^{-1} \Gamma_5\, \iota = \Gamma_5 S^{-1} \Gamma_5. \tag{11.3}$$

The solve of $D^\dagger w = r$ then goes as follows.

1. Form $r'_e = r_e - D^\dagger_{oe} (C^\dagger_o)^{-1} r_o$, so that $S^\dagger w_e = r'_e$.
2. Write $w_e = (C^\dagger_e)^{-1} z$. Then $K^\dagger z = r'_e$ with $K = C_e^{-1} S$, the operator HMC already uses.
3. Solve for z by flexible GCR with the preconditioner $C^\dagger_e\, \Gamma_5 S^{-1} \Gamma_5$. The inverse S^{-1} is one application of the cycle $\mathcal{M}$ of $\tilde{A}_R$: $S^{-1} t = C_e^{-1} \tilde{A}_R^{-1} (S_{\rm PV} C^{-1}_{{\rm PV},e})^\dagger t \approx C_e^{-1} \mathcal{M}\, (S_{\rm PV} C^{-1}_{{\rm PV},e})^\dagger t$.
4. Recover $w_e = (C^\dagger_e)^{-1} z$ and $w_o = (C^\dagger_o)^{-1} (r_o - D^\dagger_{eo} w_e)$.

Each GCR step costs one cycle, as a step of the solve with D does.

C. The system on the even sites

For completeness we derive the even-site system of $D^\dagger$. The block form of $D^\dagger$ is

$$D^\dagger = \begin{pmatrix} C^\dagger_e & D^\dagger_{oe} \\ D^\dagger_{eo} & C^\dagger_o \end{pmatrix}, \tag{11.4}$$

because the (e,o) block of $D^\dagger$ is the adjoint of the (o,e) block of D. Eliminating the odd sites of $D^\dagger w = r$ as in Sec. V gives $\big(C^\dagger_e - D^\dagger_{oe} (C^\dagger_o)^{-1} D^\dagger_{eo}\big) w_e = r'_e$, and the matrix is $S^\dagger$ term by term.

D. Why D_- cannot be inverted

Eq. (4.13) seems to offer more: the exact null vectors of $D^\dagger$, and an exact reduction of $D^\dagger w = r$ to a solve with D. Both need D_-^{-1}, and D_- cannot be inverted in practice. The reason is its spectrum. Since $D_- = cW - 1 = c\,(W - 1/c)$, D_- is c times the Wilson operator of mass $m_0 = -M_5 - 1/c$: -3 in setup S and -3.8 in setup B. The free eigenvalues of that operator,

$$m_0 + \sum_\mu (1 - \cos p_\mu) \pm i \Big(\sum_\mu \sin^2 p_\mu \Big)^{1/2}, \tag{11.5}$$

have real parts from m_0 to $m_0 + 8$, here from -3 to 5. Near real part zero they have imaginary parts of both signs, so the spectrum surrounds the origin. A Krylov method then has no low-degree polynomial p with $p(0) = 1$ that is small on the spectrum (Sec. III), and it makes no progress. On the A30 (12^4, stout, $M_5 = 1$), GCR(8) on D_- and on $D_-^\dagger$ stops at $|r|^2/|b|^2 = 0.74$ (S) and 0.96 (B) after 20000 iterations, and BiCGStab diverges. The preconditioner of Eq. (11.2) avoids D_-^{-1} altogether.

E. Measurements

Single solves. Table 3 compares three ways to solve with $D^\dagger$ from one setup of D. The right-hand sides are a random vector and the residual $r = b - D^\dagger D x_0$ after one sweep of the normal equations, solved to 10^{-10}. The second kind is the one that matters in HMC: its error lies mostly on the lowest modes.

- *The cycle of $\tilde{A}^\dagger_R$ on the same $\mathcal{P}$* (the "dual cycle"): the factorized correction (6.2) with D and $D_{\rm PV}$ exchanged.
- *Flexible GCR on $D^\dagger$ with $\Gamma_5 \mathcal{M} \Gamma_5$*, Eq. (11.2), one cycle a step.

TABLE 3. GCR iterations of a solve with $D^\dagger$ to $|r|^2/|b|^2 < 10^{-10}$ from one setup of D (32 test vectors, blocks 4^4). "rand": random right-hand side; "res": the residual after one sweep of $D^\dagger D$. "stall": not converged in 300 iterations. For comparison, the solve with D takes 7–9 (16^4) and 7–12 (12^4) iterations.

	16^4, $m = 0.01$				12^4, $m = 0.001$			
	S		B		S		B	
method	rand	res	rand	res	rand	res	rand	res
dual cycle, GCR(2)	14	67	9	13	stall	stall	12	17
dual cycle, GCR(8)	8	13	9	12	stall	stall	12	15
$\Gamma_5\mathcal{M}\Gamma_5$, GCR(2)	9	10	10	11	11	stall	12	15
$\Gamma_5\mathcal{M}\Gamma_5$, GCR(8)	9	9	10	11	10	20	11	13

The preconditioner $\Gamma_5\mathcal{M}\Gamma_5$ needs two to three iterations more than the solve with D, in both setups and on both kinds of right-hand side, as the heuristic predicts. More inner cycles per step do not reduce the count (9 iterations with one, two or four cycles on 16^4): the extra steps come from N, not from the inexact $\mathcal{M}$. The dual cycle, in contrast, stalls on the hard right-hand side in setup S.

One case is harder. On the 12^4 configuration, setup S, $m = 0.001$, the residual right-hand side needs GCR(8) (20 iterations); GCR(2) stalls. That configuration has an unpaired near-zero mode ($Q = +1$). A topological mode is localized in four dimensions, so D_- does not act on it as a number, and N need not take it out of the low modes. We have not checked this explanation. On the same right-hand side the dual cycle does not converge at all.

In HMC. On 16^4, setup S, $m = 0.01$, one short trajectory (five solves of $K^\dagger K$ to 10^{-24}) takes 313 GCR(2) iterations and 60 s with $\Gamma_5\mathcal{M}\Gamma_5$. With the dual cycle it needs GCR(8) (306 iterations, 73 s); with GCR(2), GCR(4) or GCR(6) its solves with $D^\dagger$ stagnate. The rebuild of the multigrid for new links drops from 24 to 15 s, since there is no coarse operator of $\tilde{A}_R^\dagger$ to make.

What did not work. We tried several other ways to solve with $D^\dagger$ from one setup. None is as good as $\Gamma_5\mathcal{M}\Gamma_5$.

- *The exact map.* Eq. (4.13) gives the null vectors of $D^\dagger$ as $(D_-^\dagger)^{-1}\Gamma_5 v$, and $D^\dagger w = r$ as a solve with D followed by one with $D_-^\dagger$. Both need D_-^{-1}, which does not converge (Sec. XI.D).
- *Other prolongations for the dual cycle.* $\mathcal{P}$ from $y \cup \Gamma_5 y$ converges (8–11 iterations) but doubles $\mathcal{P}$ and quadruples the coarse operators. $\mathcal{P}$ from the left singular vectors $\tilde{A}_R y$, with or without a bootstrap, or from a second setup of $\tilde{A}_R^\dagger$, needs 12–300 iterations and up to a full setup more.
- *Other coarse solvers in the dual cycle.* Its stagnation comes from the Pauli–Villars factor, solved by BiCGStab: with GCR(4) there it converges, but each cycle is three times as expensive. MINRES on the Γ_5-Hermitian form does not help, since that form is wrong at order c.

XII. Numerical results

A. Setups

Numerical setup. The gauge configurations are quenched, generated with the plaquette gauge action at $\beta = 6.0$, which corresponds to a lattice spacing of about 0.1 fm; the volumes are 12^4, 16^4, 24^4 and $24^3 \times 48$, one configuration each. On them two setups of the Möbius operator are used (Table 4), both with $(b, c) = (1.5, 0.5)$, the choice of the JLQCD and the RBC/UKQCD Collaborations, and $L_s = 8$.

TABLE 4. The two setups and the multigrid of the mass scan. All runs: Möbius $(b, c) = (1.5, 0.5)$, $L_s = 8$, $m_{\mathrm{PV}} = 1$, $\alpha_{\mathrm{PV}} = 1$, boundary $(1, 1, 1, -1)$, $\alpha = 0.6$ unless stated. Configurations: quenched, plaquette action at $\beta = 6.0$, on 12^4, 16^4, 24^4 and $24^3 \times 48$ (the 12^4 one has one unpaired near-zero mode, $Q = +1$).

	setup S (stout)	setup B (bare)
M_5	1.0	1.8
gauge smearing	stout, $\rho = 0.1$, 3 steps	none
Chebyshev smoother ν, κ_s	8, 10	12, 30
multigrid (both setups)		
aggregates	4^4 ($24^3 \times 48$: levels below with super-blocks 2, 3)	
test vectors	32 (16 filtered low modes of $\tilde{A}_R^\dagger \tilde{A}_R$, lifted to $\{s, \hat{t}\}$)	
setup	filter 2 rounds; bootstrap 2 rounds of 4 FGMRES steps	
coarse solves	single precision, $\lvert r\rvert^2/\lvert b\rvert^2 < 10^{-2}$	
precision	$\mathcal{P}$ half, cycle single, outer solver double	
outer solver	GCR(2) on $\tilde{A}_L$ to $\lvert r\rvert^2/\lvert b\rvert^2 < 10^{-18}$	

- *Setup B (bare links).* The domain-wall height $M_5 = 1.8$ is the value typical of practical simulations on unsmeared links, as adopted by the RBC/UKQCD Collaboration: on rough fields the lattice artifact of the Wilson kernel shifts the best value of M_5 above 1.
- *Setup S (stout links).* Three steps of the stout smearing of Sec. IV.E with $\rho = 0.1$ are applied to the links of the fermion operator, the setup adopted by the JLQCD Collaboration. Then $M_5 = 1.0$ is used, since the smearing largely suppresses the artifact that shifts the best value of M_5 away from 1.

The quark mass ranges over $m = 0.001$–0.1, and the Pauli–Villars mass is 1. Timings are from an NVIDIA A30; an RTX 3080 was used for iteration counts and spectral quantities only. The implementation is described in Appendix B.

What the two setups mean for the multigrid. The residual breaking of chiral symmetry at finite L_s comes from the eigenvalues near zero of the Möbius kernel $H_M = \gamma_5(b + c)W\big(2 + (b - c)W\big)^{-1}$ [8], that is from near-zero modes of $W = D_W(-M_5)$, which on rough gauge fields come from small dislocations. The two setups keep them small in two ways, smoothing the field (S) or moving M_5 (B), and so give the multigrid low modes of different character. It shows in three measured places: the smoother needs the interval $[\lambda_{\mathrm{hi}}/10, \lambda_{\mathrm{hi}}]$ in S and $[\lambda_{\mathrm{hi}}/30, \lambda_{\mathrm{hi}}]$ in B, and with B's interval setup S stagnates, with S's setup B needs nearly three times the iterations (Sec. XII.H); the field of values of $\tilde{A}_R$ reaches less far below zero with smearing ($\mathrm{Re}\, W \geq -4.6 \times 10^{-3}$ against -5.0×10^{-2} on 12^4, Sec. III); and the test vectors of $\tilde{A}_R$ are much closer to null vectors of $\tilde{A}_R^\dagger$ in S than in B ($\|\tilde{A}_R^\dagger y\|^2/\|y\|^2 \approx 0.015$ against 0.27 on 12^4 and 16^4, while $\|\tilde{A}_R y\|^2/\|y\|^2 \approx 0.003$ in both), which is why the dual cycle works in B but stalls in S (Sec. XI, Table 3).

B. Field of values: the coarse correction amplifies

The next four subsections ask why the method converges although the usual theory does not apply. Two notions from non-Hermitian linear algebra are used. The *field of values* $W(X) = \{z^\dagger X z/z^\dagger z\}$ contains the spectrum, and its leftmost point is the smallest eigenvalue of the Hermitian part $(X + X^\dagger)/2$; GCR is guaranteed to converge at every restart length when W lies in the right half-plane [23]. The *ε-pseudospectrum* is the set of z where $\sigma_{\min}(z - X) < \varepsilon$, i.e. where $\|(z - X)^{-1}\| > 1/\varepsilon$; for a normal X it is the ε-neighborhood of the spectrum, for a non-normal one it can be much larger, and it bounds the convergence of full GMRES [31].

TABLE 5. Real part of the field of values $W(X)$ and $\|I - X\|_2$ for $X = \tilde{A}_R S$, $\tilde{A}_R E$, $\tilde{A}_R \mathcal{M}$ (exact coarse solves). $m = 0.05$ on $4^3 \times 8$ (2^4 aggregates, 8 filtered low modes), $m = 0.001$ otherwise, with the multigrid of the mass scan (16 low modes, 32 test vectors, 4^4, Chebyshev of Table 4) but no levels below the coarse lattice. `preconditioned_fov_device -parts`. On 24^4 only $W(\tilde{A}_R)$ is given (Krylov dimensions 10 and 12 agree to six digits); there the field of values of $\tilde{A}_R \mathcal{M}$ did not settle (the two dimensions differ by 2%) and the eigensolvers did not fit in the memory of the A30.

	$\mathrm{Re}\, W(\tilde{A}_R)$	$\|I - \tilde{A}_R S\|$	$\|I - \tilde{A}_R E\|$	$\mathrm{Re}\, W(\tilde{A}_R \mathcal{M})$	$\|I - \tilde{A}_R \mathcal{M}\|$
$4^3 \times 8$, $M_5 = 1$	$[-5.0 \times 10^{-3}, 2.26]$		18.5	$[0.772, 1.222]$	
12^4 S	$[-4.59 \times 10^{-3}, 1.98]$	1.018	1.94×10^4	$[-23.9, 30.8]$	54.8
12^4 B	$[-5.02 \times 10^{-2}, 4.46]$	1.112	2.86×10^3	$[-1.49, 4.17]$	5.64
16^4 S			1.54×10^4	$[-15.6, 27.1]$	
16^4 B			2.5×10^3	$[-1.42, 4.27]$	
24^4 S	$[-1.31 \times 10^{-2}, 1.98]$				
24^4 B	$[-5.08 \times 10^{-2}, 4.48]$				

Why the eigenvalues alone do not decide, for readers used to Hermitian matrices: the eigenvectors of a non-normal matrix are not orthogonal, and can be nearly parallel. A vector then decomposes into eigenvectors with large coefficients of opposite sign that cancel; each component decays as its eigenvalue says, but the cancellation is lost along the way, and $\|p(X)z\|$ can grow for a while although every eigenvalue lies in the right half-plane. The field of values sees this ($z^\dagger X z$ can be negative for such a z), and so does the pseudospectrum. How nearly parallel an eigenvector is to the others is measured by the condition number of its eigenvalue, $\kappa = 1/|y^\dagger x|$ for the right eigenvector x and the left eigenvector y normalized to $\|x\| = \|y\| = 1$: $\kappa = 1$ for a normal matrix, and large κ means the eigenvalue moves by $\kappa\varepsilon$ under a perturbation of size ε, which is what widens the pseudospectrum around it (Secs. XII.C, XII.D).

The coarse correction amplifies some vectors by 10^3–10^4, and the field of values of the preconditioned operator reaches into the left half-plane (this subsection), so the bound of Eisenstat, Elman and Schultz [23] for GCR is lost. Its eigenvalues nevertheless stay at $\mathrm{Re}\,\lambda \geq 0.78$ (Sec. XII.C), and its pseudospectra bound the convergence of full GMRES (Sec. XII.D). The truncated GCR used in practice is not covered by that bound: it converges best when the coarse solves are inexact, and with accurate coarse solves it needs more iterations (setup B) or stagnates (setup S; Sec. XII.E).

The Hermitian part of $\tilde{A}_R \mathcal{M}$ was computed with $\mathcal{M}$ made a fixed linear map: both coarse factors solved to $|r|^2/|b|^2 < 10^{-20}$ in double precision with the exact adjoints, and $\mathcal{M}^\dagger$ built with the same $\mathcal{P}$ and the same smoother interval, so that $\langle z, \tilde{A}_R \mathcal{M} w\rangle$ and $\langle \mathcal{M}^\dagger \tilde{A}_R^\dagger z, w\rangle$ agree to 10^{-10}. The cycle was also taken apart: the smoother alone, $S = p(\tilde{A}_R)$, the coarse correction alone, E of Eq. (6.2), and the cycle $\mathcal{M}$ (Table 5).

On $4^3 \times 8$ the field of values of $\tilde{A}_R \mathcal{M}$ lies in the right half-plane; on 12^4 and 16^4 it does not. The field of values of $\tilde{A}_R$ itself hardly moves with the volume: from 12^4 to 24^4 its upper edge moves by less than 1% and its lower edge goes from -4.6×10^{-3} to -1.3×10^{-2} (S) and stays at -5×10^{-2} (B). The smoother is harmless ($\|I - \tilde{A}_R S\| \approx 1$); the factor that amplifies is the coarse correction, $\|I - \tilde{A}_R E\|$ of order 10^3 to 10^4 against 18.5 on $4^3 \times 8$: $\tilde{A}_R E$ is the identity on the vectors the correction is built from (Proposition 1) but amplifies other directions by three to four orders of magnitude, like a strongly oblique projection. The mechanism is the one Eick and Mawhinney [34] describe for a renormalization-group coarse lattice of Möbius fermions: the coarse inverse is large on the low modes of the coarse operator, the interpolated correction is not exactly a low mode of the fine operator, and the part that misses comes back amplified. There the fine operator is $D^\dagger D$, whose spectrum reaches 520 against 0.05 for its lowest thousand modes; it supplies the four orders of magnitude, and a Chebyshev low-pass filter of degree 150–300 is needed. Here $\tilde{A}_R$, with the Pauli–Villars operator dividing out the high modes, is close to the

TABLE 6. Eigenvalues of $\tilde{A}\mathcal{M}$ (exact coarse solves) by Krylov–Schur: the smallest real part, the condition numbers $\kappa = 1/|y^\dagger x|$ of the six eigenvalues of smallest real part (left eigenvectors from $\mathcal{M}^\dagger\tilde{A}^\dagger$), the eigenvalues farthest from 1 with their κ, and the radius about 1 of the disc that holds all the others. $m = 0.001$. `preconditioned_fov_device -arnoldi -far 1,0` (`-left` for $\tilde{A}_L$).

	$\min \mathrm{Re}\,\lambda$	κ (6 lowest)	outliers (κ)	others: $\lvert\lambda - 1\rvert \leq$
$4^3 \times 8$, $m = 0.05$, R	0.857	2–4		
$4^3 \times 8$, $m = 0.05$, L	0.854	2.8		0.156 (Ritz)
12^4 S, R	0.868	6.3–14.5	$5.86 + 0.86i$ (11.2)	0.434
12^4 S, L	0.861	7.2–26.2	$5.40 + 0.89i$ (13.9)	0.455
12^4 B, R	0.831	9.4–20.3		0.753
12^4 B, L	0.830			0.747
16^4 S, R	0.794	6.2–12.8	$10.63 + 0.19i$ (4.38)	0.474
			$3.13 + 0.02i$ (4.74)	
16^4 B, R	0.783	13.5–42.9		0.840

identity on them, and its field of values ends at 2 (S) and 4.5 (B); the large factor is the coarse inverse itself. The post-smoothing, a Chebyshev polynomial of degree 7–11 on the upper part of the spectrum, plays the role of the filter, and the cycle as a whole keeps $\|I - \tilde{A}_R\mathcal{M}\|$ at 55 (S) and 5.6 (B); how much of the remaining amplification the post-smoothing removes has not been separated. The bound of Ref. [23] for GCR therefore does not apply (Sec. III). Replacing E by the Galerkin correction of $\tilde{A}_R$ itself, $\mathcal{P}(\mathcal{P}^\dagger\tilde{A}_R\mathcal{P})^{-1}\mathcal{P}^\dagger$ (built densely), brings the field of values of setup B into the right half-plane, $[0.121, 1.99]$, and reduces $\|I - \tilde{A}_R\mathcal{M}\|$ to 1.84 (setup S: $[-3.62, 13.1]$, 16.8); but GCR(8) to 10^{-16} then needs more iterations, 16 against 13 (B) and 12 against 10 (S). The field of values is not what decides the iteration count.

C. Spectrum: far from the origin

Although the field of values reaches far into the left half-plane, every eigenvalue lies at $\mathrm{Re}\,\lambda \geq 0.78$ (Table 6), with moderate condition numbers. Setup S has one (12^4) or two (16^4) isolated outliers of large modulus that grow with the volume; all other eigenvalues lie within 0.43–0.47 of 1. The left and right forms have the same spectrum to the precision shown. At $m = 0.005$ on 16^4 the lowest real parts are 0.802 (S) and 0.790 (B).

D. Pseudospectra and a bound for GMRES

For a non-normal operator the eigenvalues alone do not bound the convergence of GMRES; the ε-pseudospectrum, the set where $\sigma_{\min}(z - X) < \varepsilon$, does [15, 31]. If a contour Γ of length L encloses the spectrum of X but not the origin and $\sigma_{\min}(z - X) \geq \varepsilon$ on it, then for any polynomial p_k with $p_k(0) = 1$,

$$\frac{\|r_k\|}{\|r_0\|} \leq \|p_k(X)\| \leq \frac{L}{2\pi\varepsilon} \max_{z\in\Gamma} |p_k(z)|. \tag{12.1}$$

An isolated outlier λ_o outside Γ is removed by the factor $(1 - z/\lambda_o)$ of p_k: then $p_k(X) = p_k(X)(I - P_o)$ with P_o the spectral projector, $\|(z - X)^{-1}(I - P_o)\| \leq \kappa_o/\varepsilon$, and with a circle $|z - c| = r$ and $p_k(z) = (1 - z/\lambda_o)(1 - z/c)^{k-1}$,

$$\frac{\|r_k\|}{\|r_0\|} \leq \kappa_o \frac{r}{\varepsilon} \frac{|\lambda_o - c| + r}{|\lambda_o|} \Big(\frac{r}{c}\Big)^{k-1}. \tag{12.2}$$

On 16^4 setup S has two outliers; the factor κ_o is then replaced by $\|I - P_1 - P_2\| \leq 1 + \kappa_1 + \kappa_2$ and there are two factors of the form $(|\lambda_o - c| + r)/|\lambda_o|$. On the real axis between the origin and the spectrum,

TABLE 7. $\min_\Gamma \sigma_{\min}(z - \tilde{A}_R\mathcal{M})$ on circles $|z - c| = r$ (16 points each, or the number given), and the bound (12.1) or (12.2): the factor per iteration r/c, the constant, and the iterations it guarantees for $\|r_k\|/\|r_0\| < 10^{-9}$. For the dense circles ε is the lower bound $\min_i \sigma_{\min}(z_i) - h/2$ on the whole circle, h the spacing of the points ($\sigma_{\min}$ is 1-Lipschitz in z); otherwise the smallest sampled value. Lanczos to 10^{-3} relative in σ^2. `preconditioned_fov_device -pseudo`.

	c	r	$\min \sigma_{\min}$	r/c	constant	k for 10^{-9}
12^4 S	1	0.6	0.0285	0.6	220	53
12^4 S, 160 points	1	0.6	0.0281, ≥ 0.0164	0.6	380	54
12^4 S	1	0.5	0.0107	0.5	480	40
12^4 B	1.4	0.93	0.0985	0.66	9.4	57
12^4 B, 96 points	1.4	0.93	0.0980, ≥ 0.0676	0.66	13.8	58
12^4 B	1.3	0.87	0.0860	0.67	10.1	58
16^4 S	1	0.6	0.0262	0.6	190	53
16^4 S	1	0.55	0.0135	0.55	340	47
16^4 B	1.4	1.05	0.1295	0.75	8.1	80

$\sigma_{\min}(z - \tilde{A}_R\mathcal{M}) \geq 0.024$ (S) and ≥ 0.031 (B) on 12^4 ($z = -0.5 \ldots 0.7$), and 0.107 (S), 0.271 (B) at the origin.

The bound guarantees convergence at a fixed rate, r/c per iteration, without stagnation; it is pessimistic by a factor of 3–5 in iterations (GCR needs 10–17 with exact coarse solves, Sec. XII.E), and its rate is that measured on $24^3 \times 48$, where the median factor per GCR step of $|r|^2$ is 0.23–0.34 (Table 9), i.e. 0.48–0.58 in $|r|$, against $r/c = 0.5$–0.6 for setup S. Setup B is not worse between 12^4 and 16^4 ($\min \sigma$ grows from 0.098 to 0.13); in setup S the outliers grow but their condition numbers fall and ε stays the same.

What is not yet rigorous. (i) With 16 points per circle the spacing is larger than ε, and the sampled minimum says nothing about the points in between. On the two dense circles of 12^4 the spacing is below ε, and since $\sigma_{\min}(z - X)$ is 1-Lipschitz in z the bound holds on the whole circle, $\sigma_{\min} \geq 0.0164$ (S) and ≥ 0.0676 (B); the guaranteed iterations change from 53 to 54 and from 57 to 58. The circles of 16^4 have only the 16 points. (ii) Lanczos approaches $\sigma^2_{\min}$ from above; a rigorous lower bound needs an inclusion theorem (Kato–Temple) with a bound on the second singular value. (iii) That the disc holds all eigenvalues but the outliers rests on the four eigenvalues farthest from c found by Arnoldi. (iv) The bound is for full GMRES with a fixed linear $\mathcal{M}$ (Sec. XII.E). (v) The spectra, and the FOV of $\tilde{A}_R\mathcal{M}$, are not available on 24^4 and $24^3 \times 48$: the eigensolvers do not fit in the memory of the A30. This is a limit of the hardware used here, not of the method.

E. Truncated GCR: inexact coarse solves

With the usual coarse solves to 10^{-2}, GCR(2) needs one to five iterations more than the full method (Table 8). With accurate coarse solves, where $\mathcal{M}$ is close to the fixed map of Secs. XII.B–XII.D, GCR(2) stagnates ($|r|^2 \approx 3 \times 10^{-8}$ after 1000 iterations) while GCR(8) converges in 26 and GCR(≥ 16) in 17. This is what the analysis predicts: the pseudospectra guarantee the full method; a truncated or restarted one has no such guarantee when the field of values contains the origin [23, 32]. That the truncated solver works with inexact coarse solves is an empirical fact, and the obvious explanation does not hold: a coarse solve stopped at 10^{-2} does not leave out the directions in which E amplifies. On the vector x at the left end of the field of values of $\tilde{A}_R\mathcal{M}$ (exact coarse solves; 12^4, `preconditioned_fov_device -parts`), the cycle with coarse solves to 10^{-2} gives $\mathrm{Re}\, x^\dagger \tilde{A}_R\mathcal{M}x = -25.2$ (S) and -1.43 (B), against -24.3 and -1.49 with exact ones, and $\|(I - \tilde{A}_R E)x\| = 1.16 \times 10^3$ and 1.03×10^2, against 1.13×10^3 and

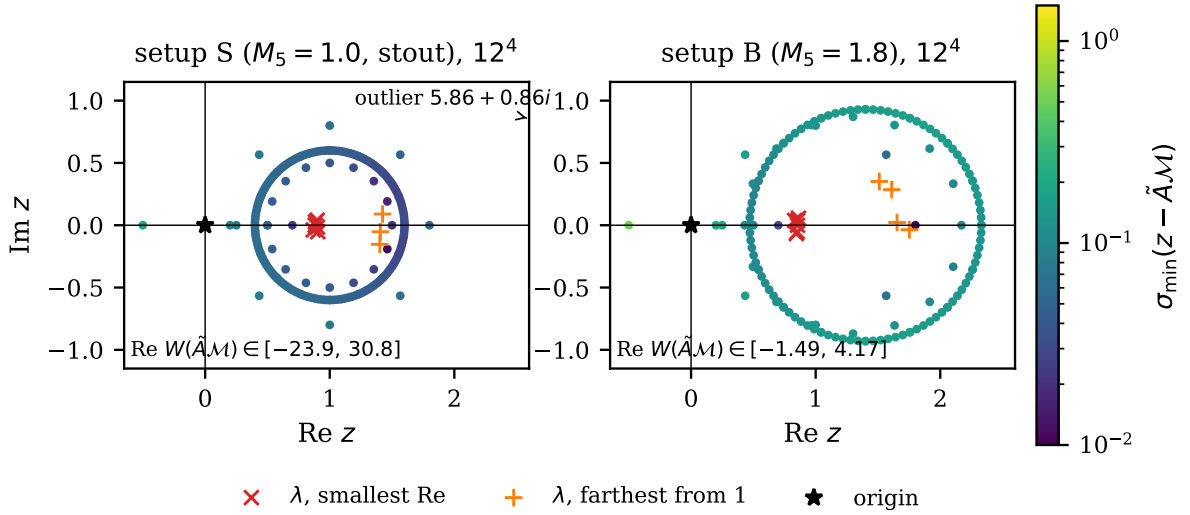


FIG. 5. 12^4, $m = 0.001$, exact coarse solves: the six eigenvalues of $\tilde{A}_R\mathcal{M}$ of smallest real part, the four farthest from 1, and $\sigma_{\min}(z - \tilde{A}_R\mathcal{M})$ (color) at the points where it was computed: the real axis, the contours of Table 7 and the dense circles $|z - 1| = 0.6$ (S) and $|z - 1.4| = 0.93$ (B). The field of values reaches far into the left half-plane (its real range in the corner), the spectrum does not. `data/pseudo_*.tsv`, `data/spectrum_*.tsv`.

TABLE 8. GCR(k) iterations to $|r|^2/|b|^2 < 10^{-18}$, 12^4, $m = 0.001$, the multigrid of Table 4 (no levels below the coarse lattice), one random right-hand side; the point source: range over its 12 columns.

	coarse solves	$k = 2$	8	16	32	64
S, $\tilde{A}_L$	10^{-2}	14			12	
S, $\tilde{A}_R$	10^{-2}	12			11	
B, $\tilde{A}_L$	10^{-2}	14			13	
B, $\tilde{A}_R$	10^{-2}	15			14	
S, $\tilde{A}_L$, point source	10^{-2}	15–19			13–14	
S, $\tilde{A}_L$	10^{-8}	no conv.	26	17	17	17
S, $\tilde{A}_L$	10^{-4}	18			16	

1.12×10^2; with tolerances 10^{-8} and 10^{-4} they are those of the exact solves to within 1%. What the tolerance changes is that the cycle is no longer a fixed linear map: with Krylov coarse solves stopped early it depends on its input. On $24^3 \times 48$ the full method cannot be run on the A30: GCR(8) and GCR(16) run out of device memory after the setup (GCR(2) peaks at 19.9 of 24 GB, and every further direction is another fine-grid vector).

No run shows a plateau (Table 9, Fig. 6); the first step reduces $|r|^2$ by about 10^{-5} in every run.

What this establishes. As is usual for multigrid preconditioners of lattice Dirac operators, the convergence of the truncated flexible outer method is established numerically: GCR(2) with the coarse solves stopped at 10^{-2} converged in every run of this section, without plateaus, at the rate that the pseudospectral bound of Sec. XII.D gives for the full method. A guarantee for a truncated method whose preconditioner varies from step to step is not available for flexible Krylov methods with multigrid preconditioners in general. That accurate coarse solves make GCR(2) slower, and in setup S stagnate, is an observation of its own; since the loose solves do not avoid the amplifying directions, the variation of the cycle with its input is the candidate, which a fixed but inaccurate linear coarse solve, or the exact cycle perturbed at random, would test.

TABLE 9. Residual history of GCR(2), $24^3 \times 48$, $\tilde{A}_L$, the multigrid of Table 4: steps, median and largest factor of $|r|^2$ per step, and the longest stretch of steps with a factor above $1/2$. From an earlier build than the mass scan; the steps differ from those of Table 10 (`6d5bf18`) by at most one.

setup	m, α	steps	median	worst	slow stretch
S	0.001, 0.6	28	0.315	0.564	2 steps
S	0.001, 1.0	27	0.338	0.479	none
S	0.002, 0.6	23	0.233	0.426	none
S	0.002, 1.0	23	0.253	0.354	none
B	0.001, 0.6	26	0.299	0.357	none
B	0.001, 1.0	28	0.317	0.403	none
B	0.002, 0.6	23	0.246	0.300	none
B	0.002, 1.0	25	0.280	0.347	none

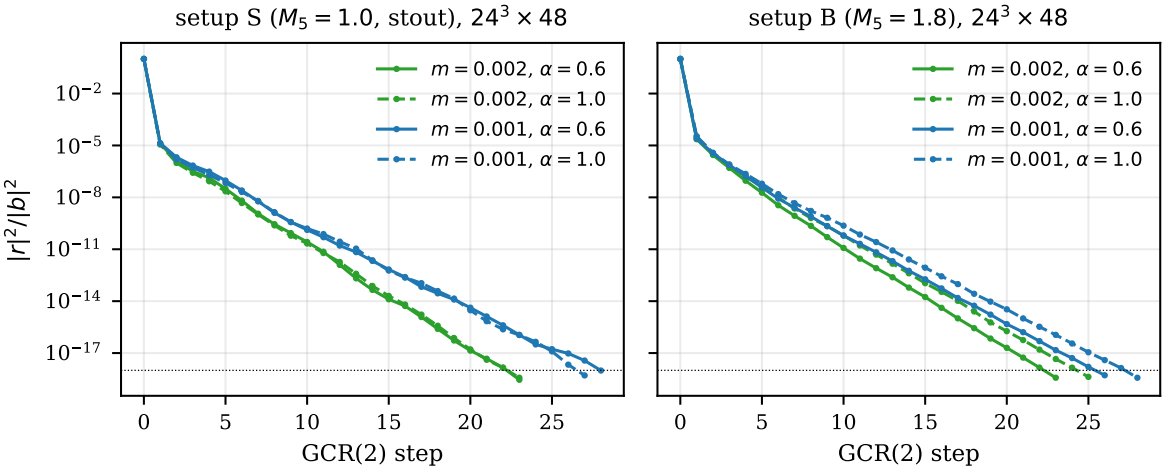


FIG. 6. The GCR(2) residual histories of Table 9, $24^3 \times 48$. `data/gcr_steps_24.tsv`.

F. Comparison with CGNE

What the multigrid costs, against the reference solver of Sec. X, on $24^3 \times 48$: one random right-hand side over seven masses, then whole point-source propagators.

Table 10 and Fig. 7 are the mass scan on $24^3 \times 48$. The CGNE iterations grow by a factor of 21 (S) and 18 (B) from $m = 0.1$ to $m = 0.001$, the multigrid iterations by 2.2 (S) and 3.7 (B). At $m = 0.001$ the multigrid solve is 7.5 times (S) and 9.0 times (B) faster than the double-precision CGNE; with the setup of about 210 s it pays for itself after 2–3 right-hand sides, i.e. within one point-source propagator (12 columns) for $m \leq 0.01$. At $m = 0.1$ in setup S the multigrid is not faster at all. The multigrid needs three times the device memory of CGNE; on the 24 GB A30 this excludes GCR(k) with $k \geq 8$ (out of memory after the setup). Neff's $\alpha = 0.6$ against $\alpha = 1$ in the left form: equal within one iteration for S except at $m = 0.1$ (three fewer), two to three iterations fewer for B. Against an earlier build (`6c7ff42`) the iteration counts differ by one or two in six of the 28 multigrid runs; the cause (code change or non-determinism of the device reductions) is not established.

Mixed-precision CGNE. For a comparison on equal terms the reference solver also runs in mixed precision: CG in single precision on the normal equation of $C_e^{-1}S$, with reliable updates in double precision (the true residual recomputed whenever $|r|^2$ has fallen by 10^{-2}). It needs the iterations of double precision (at $m = 0.001$ in setup S 14% more) and 0.54–0.61 times the time, with the same final residuals

TABLE 10. Mass scan on $24^3 \times 48$, A30, one random right-hand side to $|r|^2/|b|^2 < 10^{-18}$ on the even-odd system: even-odd CGNE in double and in mixed precision against the multigrid (GCR(2) on $\tilde{A}_L$), $\alpha = 0.6$ for both and $\alpha = 1$ for the multigrid; iterations, solve time in seconds, coarse-operator applications, and N_{rhs}, the number of right-hand sides at which the multigrid setup (200–237 s) is paid back, $t_{\mathrm{setup}}/(t_{\mathrm{CGNE}} - t_{\mathrm{MG}})$, against the double- and the mixed-precision CGNE. The mixed-precision CGNE needs the iterations of the double-precision one, except at $m = 0.001$ (S): 4090. Final $|D\psi - \eta|^2/|\eta|^2$: 9×10^{-18}–3×10^{-16} (CGNE, both), 2×10^{-19}–3×10^{-18} (MG). Device memory peak: CGNE 6.9 GB (double) and 7.2 GB (mixed), MG 19.9 GB ($\alpha = 0.6$). RutticeQCD `6d5bf18`; `data/massscan_24.tsv`.

	CGNE			MG, $\alpha = 0.6$			MG, $\alpha = 1$		N_{rhs}	
m	it	s	s, mixed	it	s	coarse	it	s	double	mixed
setup S ($M_5 = 1.0$, stout)										
0.1	173	4.8	2.9	13	5.9	1202	16	6.8	—	—
0.05	291	8.1	4.6	13	5.3	750	14	5.7	72	—
0.02	644	17.7	9.7	14	5.7	830	14	5.7	17	51
0.01	1166	32.0	17.4	15	6.2	990	15	6.3	7.9	18
0.005	2067	56.6	30.5	17	7.1	1158	17	7.3	4.2	8.8
0.002	3419	93.5	50.2	23	10.1	1954	22	10.1	2.5	5.1
0.001	3602	98.5	59.9	28	13.2	3014	27	13.1	2.4	4.4
setup B ($M_5 = 1.8$)										
0.1	231	6.4	3.7	7	3.3	201	9	4.2	69	535
0.05	402	11.1	6.2	10	4.7	302	12	5.7	33	143
0.02	857	23.6	12.9	13	6.2	424	15	7.2	12	32
0.01	1447	39.6	21.4	15	7.2	556	17	8.2	6.6	15
0.005	2299	63.0	34.0	18	8.7	714	21	10.2	4.0	8.6
0.002	3473	95.2	51.2	23	11.3	1024	25	12.4	2.8	5.8
0.001	4228	115.6	62.0	26	12.9	1274	28	13.9	2.1	4.3

and 7.2 GB of device memory. Against it the multigrid solve at $m = 0.001$ is 4.5 (S) and 4.8 (B) times faster, and the setup is paid back after 4–5 right-hand sides, within one propagator for $m \leq 0.005$; for $m \geq 0.05$ (S) the mixed-precision CGNE solve is the faster one, and at $m = 0.1$ (B) the two take nearly the same time.

A point-source propagator. The break-even estimates are confirmed by whole propagators, the 12 columns of a point source (Table 11). At $m = 0.001$ the multigrid takes half the wall time of the mixed-precision CGNE including its setup, the solve per column being 4.5–5 times faster; at $m = 0.01$ the setup is not quite paid back within the 12 columns, as $N_{\mathrm{rhs}} \approx 15$–18 (Table 10) predicts. A point source needs more iterations than a random right-hand side, with either solver. The pseudoscalar correlators of the two agree to 4×10^{-8}–7×10^{-7} at $t \leq 3$ and to 2×10^{-6} at all 48 times. The multigrid stops on the residual of D ($|D\psi - \eta|^2/|\eta|^2$ of 1–2 $\times 10^{-18}$), the CGNE on that of its normal equation (6×10^{-17} to 3×10^{-16}). The device memory is 21.8 GB (multigrid) and 9.1 GB (mixed-precision CGNE).

G. Γ_5-Hermiticity for $c \neq 0$

For $c = 0$ the coarse operators are Σ-Hermitian to rounding; for $c \neq 0$ they are not, by a few percent in random directions and up to 14% in the directions of the smallest singular values, growing linearly with c (Table 12), as Eq. (4.10) says. This does not hurt the method: with MINRES on $\mathcal{H}_c$ for the Pauli–Villars factor the outer iterations at $c = 0.5$ are those of the Shamir operator ($c = 0$, a different

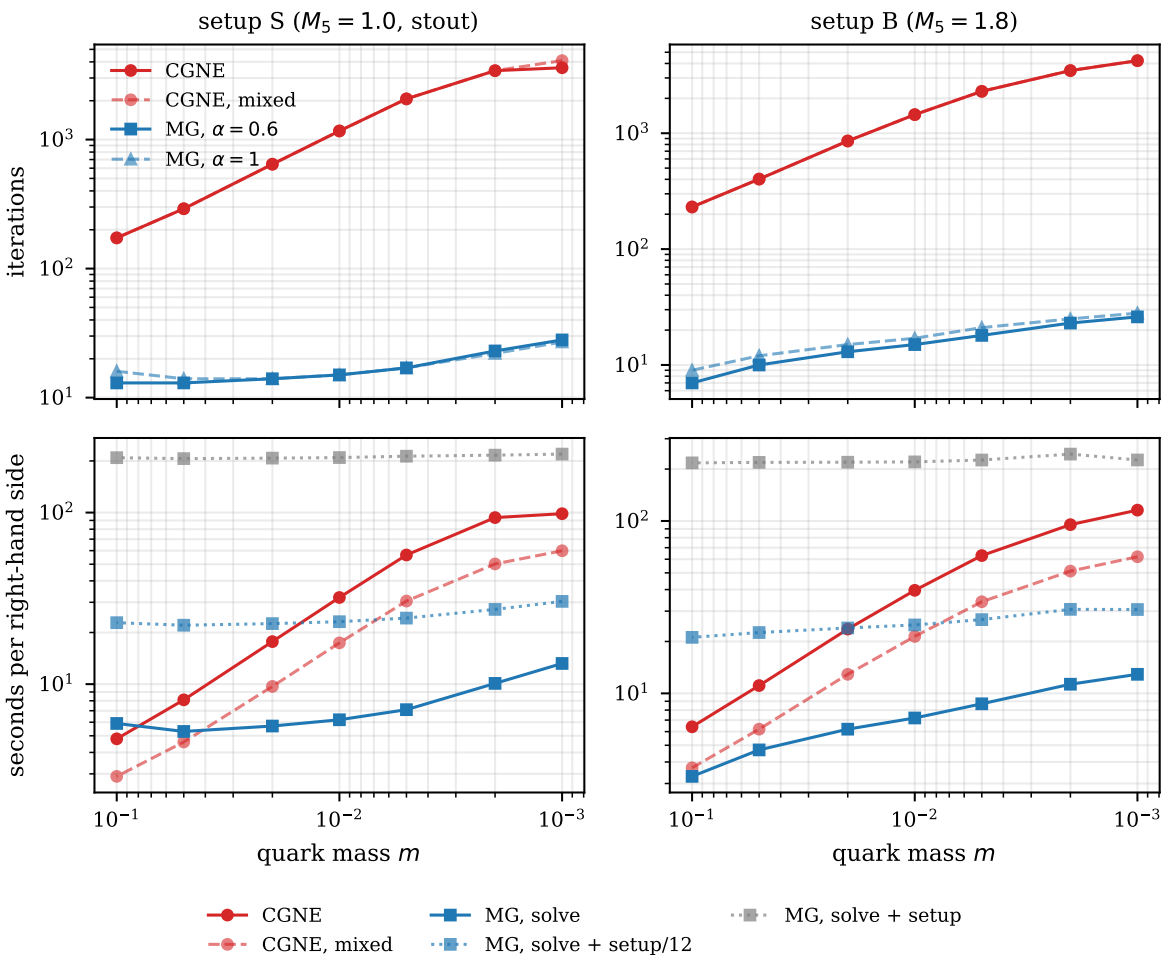


FIG. 7. The mass scan of Table 10 and the mixed-precision CGNE: iterations (top) and seconds per right-hand side (bottom) against the quark mass, $24^3 \times 48$, A30. "Solve + setup/12": the setup spread over the 12 columns of one point-source propagator. `data/massscan_24.tsv`, `figures.py`.

operator, for which MINRES is exact), and MINRES is better than BiCGStab on the exact adjoint in every case (Table 13); with BiCGStab the stout setup nearly stagnates. The non-Hermitian part of $\mathcal{H}_c$ is small enough for MINRES stopped at 10^{-2}; why BiCGStab is so much worse, also at $c = 0$, has not been studied.

H. Sensitivity to the parameters

The method is robust within a window of each parameter and fails outside it (Table 14). The smoother interval must suit the setup; the bootstrap is necessary; more accurate coarse solves are not better, and with GCR(2) the most accurate ones stagnate in setup S (Sec. XII.E). Larger setups (smaller aggregates, more test vectors) buy fewer iterations, but on 12^4 not less time (Table 15). All runs of the table use GCR(2); the stagnations are of the kind of Sec. XII.E, and whether full GCR would remove them has not been tested. Lüscher's test of local coherence for these test vectors has not been made.

Fewer iterations are not always less time (Table 15): 2^4 aggregates need 8–9 iterations but a larger coarse lattice, so the solve and the setup take about twice as long as with 4^4; 48 test vectors save a third of the iterations and lose in setup and in time per iteration. For Wilson fermions, Whyte, Stathopoulos and Romero [35] obtain the effect of more test vectors at the coarse size of fewer: they generate more, truncate them by a singular value decomposition on each aggregate and chirality (as $\mathcal{P}$ is split here by Γ_5), and stream the decomposition to save device memory. Applied here, the truncated basis would no longer

TABLE 11. Point-source propagators (12 columns) on $24^3 \times 48$, A30: the multigrid (setup seconds, iterations and seconds per column) against the mixed-precision CGNE (iterations and seconds per column), the wall time of the whole propagator, and the largest relative difference of the pseudoscalar correlator at $t = 0 \ldots 3$. RutticeQCD `6d5bf18`; `data/pointsource_24.tsv`.

		MG			CGNE, mixed		wall (s)		
	m	setup	it	s	it	s	MG	CGNE	$\lvert\Delta C\rvert/C$
S	0.001	206	29–32	13.6–14.9	4182–4572	61.4–67.0	396	810	9×10^{-8}
B	0.001	215	29–30	14.2–14.8	5023–5332	73.5–78.0	409	918	7×10^{-7}
S	0.01	203	16–17	6.4–7.0	1337–1388	19.8–20.5	305	264	2×10^{-7}
B	0.01	215	17	8.1	1641–1745	24.2–25.7	331	324	4×10^{-8}

TABLE 12. $\delta(v) = \|\Sigma X \Sigma v - X^\dagger v\|/\|X^\dagger v\|$ for the coarse operators $X = \mathcal{D}_\mathrm{c} = \mathcal{P}^\dagger D \mathcal{P}$ and $\mathcal{P}^\dagger D_\mathrm{PV} \mathcal{P}$, $X^\dagger$ the exact adjoint, Σ the coarse Γ_5; $b - c = 1$ throughout (as usual for Möbius fermions), $L_s = 8$, $m = 0.001$ ($4^3 \times 8$: $m = 0.05$, 2^4 aggregates, 32 coarse components; 12^4: 4^4, 64). Random v (four) and the right singular vectors of the four smallest singular values. `examples/coarse_hermiticity`. *Lanczos on $X^\dagger X$ not converged in 2000 restarts (the low end of $\mathcal{D}_\mathrm{c}$ is crowded): approximate vectors.

	$\mathcal{P}^\dagger D \mathcal{P}$		$\mathcal{P}^\dagger D_\mathrm{PV} \mathcal{P}$	
	random	low	random	low
$4^3 \times 8$, $M_5 = 1$, $(b, c) = (1, 0)$	3.8×10^{-16}	9×10^{-16}	4.0×10^{-16}	9×10^{-16}
$4^3 \times 8$, $M_5 = 1$, $(b, c) = (1.5, 0.5)$	0.041	0.065–0.074	0.050	0.090–0.095
12^4 S, $(b, c) = (1.25, 0.25)$	0.012	0.034–0.050*	0.030	0.023–0.031
12^4 S, $(b, c) = (1.5, 0.5)$	0.023	0.067–0.099*	0.053	0.041–0.054
12^4 B, $(b, c) = (1.5, 0.5)$	0.030	0.140–0.146	0.055	0.053–0.066

contain s and $\hat{t}$ exactly, so the exactness of Proposition 1 would be lost; whether the gain outweighs that has not been tried. With the setup as fast as it is now, the base setting is no longer the time optimum on 12^4: 16 test vectors (a coarse dimension of 32) halve the setup at a solve that is no slower, and more smoother steps ($\nu = 12$, 16 for setup S) solve faster at the same setup. On $24^3 \times 48$ the fewer test vectors do not carry over (setup S, $m = 0.001$, point source, A30, `6d5bf18`): 16 halve the setup there too (116 against 207 s), but the columns that converge need 910–940 iterations and the others stop at the cap of 2000; 24 (184 s of setup) need 162 iterations for the first column, against 29–32 with 32. More smoother steps have not been tried there. All runs use one setup seed, and the iterations move by a few between builds that add in other orders, so differences of a few iterations are within that variation.

XIII. Open problems

The following questions remain open. Those the measurements answer are discussed where they arise: the left form in Sec. VI.C, $c \neq 0$ in Secs. VII and XII.G, the field of values in Secs. III and XII.B–XII.D.

1. **Choosing the smoother interval.** The interval $[\lambda_\mathrm{hi}/\kappa_s, \lambda_\mathrm{hi}]$ is set from an estimate of the spectral radius and a ratio chosen per setup, and the wrong ratio makes the solve stagnate (Sec. XII.H); the setup works with singular values of $\tilde{A}_R$, which for a non-normal operator need not match its eigenvalues. How to choose the interval from the operator, which part of the low end is left to the smoother after the coarse correction, and whether an ellipse with complex foci should replace the real interval, are open.

TABLE 13. GCR(2) iterations (coarse applications) with the Pauli–Villars factor solved by MINRES on $\mathcal{H}_c$, which assumes $\delta = 0$, or by BiCGStab on the exact adjoint, both to 10^{-2}; 12^4, $m = 0.001$, $\tilde{A}_L$, otherwise Table 4.

	$c = 0.5$, $b = 1.5$		$c = 0$, $b = 1$	
	MINRES	BiCGStab	MINRES	BiCGStab
setup S	14 (3098)	373 (192892)	13 (3061)	24 (7276)
setup B	14 (2351)	19 (3876)	15 (2103)	28 (3540)

TABLE 14. GCR(2) iterations when one parameter of Table 4 is changed, 12^4, $m = 0.001$, $\tilde{A}_L$, one random right-hand side (base: 16 in setup S, 14 in setup B); "stag.": not converged in 1000 iterations. Setup times in seconds where they change much. RTX 3080, RutticeQCD `6d5bf18`; `data/sensitivity_12.tsv`.

change	setup S	setup B
aggregates 2^4 / 3^4 / 6^4	8 (22 s) / 12 / 23	9 (15 s) / 11 / 21
test vectors 8 / 16 / 24 / 48	stag. / 21 / 26 / 10 (17 s)	38 / 18 / 15 / 11 (14 s)
smoother $\nu = 4$ / 6 / 8 / 12 / 16	28 / 18 / 16 / 11 / 9	stag. / 96 / 27 / 14 / 13
interval $\kappa_s = 5$ / 10 / 30 / 100	23 / 16 / stag. / stag.	stag. / 38 / 14 / stag.
coarse solves 10^{-1} / 10^{-2} / 10^{-4} / 10^{-8}	23 / 16 / 18 / stag.	20 / 14 / 16 / 17
bootstrap rounds 0 / 1 / 2 / 3	stag. / 15 / 16 / 22	249 / 46 / 14 / 12
right form $\tilde{A}_R$	11	15

2. **Even-odd smoother and coarse space.** Boyle and Yamaguchi [20] found that a red-black smoother combined with the HDCR coarse correction was not effective, and attributed it to the eigenvectors of the red-black operator and of $\gamma_5 R_5 D$ not being aligned. In the present method the smoother and the correction both approximate $\tilde{A}^{-1}$ and the coarse space is built from low modes of $\tilde{A}_R$, so that particular mismatch is absent by construction; but the coarse operators are Galerkin projections of D and D_{PV}, not of $\tilde{A}$. Whether the even-odd outer system is preferable to the full-lattice A' for this method has not been established.

3. **A rigorous bound, and larger volumes.** The pseudospectral bound rests on Lanczos estimates of $\sigma_{\min}$, which approach it from above, and on the eigenvalues farthest from the centre found by Arnoldi (Sec. XII.D). A bound from an inclusion theorem, and how the outliers of setup S and the pseudospectra develop beyond 16^4, are open. The latter is only a matter of memory: on 24^4 the field of values of $\tilde{A}_R$ fits in the 24 GB of the A30 but the Arnoldi iteration does not. A GPU with more memory would answer it with the programs of Sec. B unchanged; so would the CPU, with enough memory, once the Arnoldi and pseudospectrum options are carried over to the CPU version of the program.

4. **How many test vectors.** Local coherence suggests that the number of test vectors an aggregate needs does not grow with the volume, yet 16 suffice on 12^4 and on $24^3 \times 48$ even 24 do not (Sec. XII.H). Whether the 16 lowest modes of a larger lattice cover too narrow a part of the spectrum, or whether the low modes vary in the fifth dimension more than one vector per aggregate can follow (Eick and Mawhinney [34] find the fifth-dimensional profiles of low modes to differ between lattices), is open; a test vector split in s as it is now split in chirality would keep s and $\hat{t}$ in the range of $\mathcal{P}$, and Lüscher's coherence test would measure what the coarse space captures.

5. **Solving with $D^\dagger$: the topological mode.** With $\Gamma_5 \mathcal{M} \Gamma_5$ (Sec. XI) the solves with $D^\dagger$ take two or three iterations more than those with D, except on the 12^4 configuration with an unpaired

TABLE 15. The changes of Table 14 that move the cost, timed on the A30: iterations / setup seconds / solve seconds. 12^4, $m = 0.001$, $\tilde{A}_L$, GCR(2), one random right-hand side, RutticeQCD `6d5bf18`; `data/costs_12.tsv`. The iterations are those of Table 14 (RTX 3080, the same build) in every row; with an earlier build they differed by up to six, the device kernels adding in other orders.

change	setup S	setup B
base	16 / 8 / 0.7	14 / 7 / 0.5
aggregates 2^4	8 / 18 / 1.4	9 / 12 / 1.2
aggregates 3^4	12 / 8 / 0.5	11 / 7 / 0.5
aggregates 6^4	23 / 10 / 0.6	21 / 10 / 0.6
test vectors 16	21 / 4 / 0.5	18 / 4 / 0.4
test vectors 24	26 / 6 / 0.9	15 / 6 / 0.4
test vectors 48	10 / 16 / 0.7	11 / 13 / 0.8
smoother $\nu = 4$	28 / 8 / 1.2	stag.
smoother $\nu = 6$	18 / 7 / 0.6	96 / 6 / 3.1
smoother $\nu = 12$	11 / 8 / 0.5	(base)
smoother $\nu = 16$	9 / 8 / 0.4	13 / 7 / 0.5
bootstrap rounds 1	15 / 5 / 0.5	46 / 5 / 1.1
bootstrap rounds 3	22 / 10 / 1.0	12 / 10 / 0.5

near-zero mode, where the residual right-hand side of HMC needs GCR(8). Whether the localized topological mode is the cause, and whether deflating it lets GCR(2) converge there, is open. Why $N = c[Y, W]D^{-1}$ has a small square, beyond the heuristic of Sec. XI, is open as well.

TABLE 16. The notation of this note (and of RutticeQCD) and that of other references. Equation numbers are those of the references; "the same" means the same operator.

	This note	Ref. [8]	Ref. [24]	Bridge++ code
Height	$M_5 > 0$	$-M_5$, $M_5 = O(-1)$ (2.13)	M_0	M_0, `domain_wall_height`
Fifth dim.	$s = 0, \ldots, L_s - 1$	$s = 1, \ldots, L_s$	$s = 1, \ldots, N_s$	$0, \ldots, N_s - 1$, `extent_of_5th_dimension`
Wilson part	$W = (4 - M_5) - \frac{1}{2}K$	$D^{\mathrm{Wilson}}(M_5)$ (2.13)	$D_{\mathrm{W}}(-M_0)$	$\frac{1}{2\kappa}D_{\mathrm{W}}$, $\kappa = 1/(8 - 2M_0)$
$D_\pm$	$D_+ = 1 + bW$, $D_- = cW - 1$ (Eq. (4.3))	$D_+ = bD^{\mathrm{W}} + 1$, $D_- = cD^{\mathrm{W}} - 1$ (2.18)	the same (2)	—
D	Eq. (4.7)	$D_{\mathrm{DW}}(m)$ (2.20)	$D(m)$ (1)	`Fopr_Domainwall`, mode `D`
C	$d_+ + d_- Y$, Eq. (4.14)	—	$\mathrm{diag}(D_{ee}, D_{oo})$ (4)	D_{ee}, D_{oo} of `_eo`; `m_dp` $= \alpha d_+$, `m_dm` $= \alpha d_-$
Neff's α	DQ, Eq. (4.21)	— (its $\alpha = \frac{b+c}{b-c}$)	—	`parameter_alpha`: DQ, site part CQ
Γ_5	$\gamma_5 R$	$\gamma_5 R D_-^{-1}$ (2.22)	—	—
4d quark	$P_-\psi_0 + P_+\psi_{L_s-1}$	$P_-\Psi_1$, $P_+\Psi_{L_s}$	—	the same

Appendices

A. The notation of other references

Table 16 relates Eqs. (4.6)–(4.21) to Brower, Neff and Orginos [8], to Kanamori, Chen and Matsufuru [24] and to the code of Bridge++ [14] (`Fopr_Domainwall`, `Fopr_Domainwall_eo`, with Neff's α as `parameter_alpha`). Eq. (4.4) is the tridiagonal matrix of Ref. [8], Eq. (2.20), and of Ref. [24], Eq. (1), with $D_+ = 1 + bW$ on the diagonal, $D_- P_+$ below and $D_- P_-$ above it, $-mD_- P_\pm$ in the corners, and s shifted by one. The operators D, C and $D^{(\alpha)}$ are thus the same in all four; the differences are names and signs:

- $D_\pm$ are those of Refs. [8, 24], $D_- = cW - 1$. The hop Y of Eq. (4.6) carries a minus sign, so $D = D_+ - D_- Y$ and $d_- = 1 - c(4 - M_5)$ is minus the site part of D_-. The comments of RutticeQCD on the five-dimensional source call $1 - cW$ "D_-", which is $-D_-$ (the source is that of the Bridge++ domain-wall test, sign included).

- The height $M_5 > 0$ here is M_0 in Ref. [24] and in Bridge++ (`domain_wall_height`); Ref. [8] writes M_5 for the Wilson mass, $M_5^{\mathrm{BNO}} = -M_5$, Eq. (2.13). The caption of Fig. 1 of Ref. [24] gives $M_0 = -1$ for the height 1 of its text.

- The α of Ref. [8], $(b + c)/(b - c)$, is the scale factor of the Möbius kernel (2 at $b = 1.5$, $c = 0.5$), not Neff's α of Eq. (4.21).

- $\Gamma_5 = \gamma_5 R D_-^{-1}$ in Ref. [8], Eq. (2.22); here $\Gamma_5 = \gamma_5 R$.

$D^{(\alpha)}$ here and the operator of Bridge++ with `parameter_alpha` were compared on a 12^4 configuration with stout links, $L_s = 8$, $m = 0.01$: at $\alpha = 0.6$ and $\alpha = 1$ CG on the even-odd normal equation took the same number of iterations for every column of a point source, and the pseudoscalar correlators agree to 10^{-11}. Bridge++ returns $\psi^{(\alpha)}$; $\psi = Q\psi^{(\alpha)}$.

B. Implementation and reproduction

Implementation. The method is implemented in RutticeQCD (Rust), on the CPU and on NVIDIA GPUs through CUDA kernels of the Wilson and Möbius operators taken from Bridge++. On the device the fine operators, the prolongation, the coarse operators and the coarse Krylov solvers all stay on the GPU. The precisions are those of Table 4: the prolongation is stored in half precision unless asked otherwise, the cycle above the coarse lattice runs in single precision on operators converted on the device from the links of the double-precision ones, and the outer solver is in double precision. The coarse solvers are BiCGStab for the quark factor and MINRES on $\mathcal{H}_c$ for the Pauli–Villars factor (Sec. VII), both stopped at $|r|^2/|b|^2 < 10^{-2}$ unless stated, on the even coarse sites. The numbers of Sec. XII come from the parameter-file program `ruttice` and from the examples `preconditioned_fov` (CPU) and `preconditioned_fov_device` (GPU), whose options are given with each result; the remarks after Proposition 1 from the example `correction_exactness` (CPU). The numbers of Sec. XI come from the examples `mobius_gamma5_check` (Eqs. (4.10) and (4.13), CPU), `dminus_cost` (the inverse of D_-, GPU), `dual_basis` (prolongations for the dual cycle, GPU) and `dagger_precond` (Table 3, GPU). In HMC the solves with $D^\dagger$ use $\Gamma_5\mathcal{M}\Gamma_5$ (`fermion::adjoint_by_gamma5`). The eigensolvers (thick-restart Lanczos, Krylov–Schur Arnoldi [30] with left eigenvectors, and $\sigma_{\min}(z-X)$ by Lanczos on $(z-X)^\dagger(z-X)$) are in the module `eigen` and are checked against Bridge++'s eigensolver test.

All numbers of Sec. XII come from one configuration per volume; the most useful extensions are more configurations, a larger volume and dynamical ensembles. The rest of this appendix gives what is needed to run them.

Code. RutticeQCD, `https://github.com/atsanroger/RutticeQCD`, branch `cuda-bridge` (the commit of each result is given with it; later commits give the same iterations, up to the two-iteration differences between builds noted in Sec. XII.F). It needs Rust (stable) and, for the GPU, CUDA 12:

```
export CUDA_HOME=/usr/local/cuda
cargo build --release --features cuda-bridge
./target/release/ruttice mg_24.yaml
```

The GPU backend runs on one GPU. The parameter files are in the style of Bridge++ and are documented in `docs/USAGE.md`; gauge configurations are read in the Bridge++ text and binary formats, NERSC, ILDG and SciDAC.

The mass scan (Table 10, Fig. 7). The parameter files are `docs/mg_background/runs/mg_24.yaml` (the multigrid) and `runs/cgne_24.yaml` (the reference, double or mixed precision); the comments in them say how to switch to setup B and to a point source. Each run prints a line

```
seed 91: GCR 28 iterations on the device; 3014 coarse applications;
    worst |Dx-b|^2 2e-18; 13.2 s; ... device 19.91 GB
```

and, for the multigrid, `setup on the device:  206 s`. These, per mass and setup, are what Table 10 lists. On an A30 one configuration takes about 50 minutes for the multigrid (seven masses, both setups, $\alpha = 0.6$; the setup dominates) and under half an hour for the two CGNE scans.

Memory and volume. On $24^3 \times 48$ the multigrid needs 19.9 GB of device memory (CGNE 6.9 GB in double, 7.2 GB mixed), roughly proportional to the volume: $32^3 \times 64$ needs about 63 GB (an 80 GB GPU), $48^3 \times 96$ more than one GPU has. The aggregates, 4^4, must divide the lattice, and each level below the coarse lattice must divide the extent of the level above (`MultiGrid_Level2`, `_Level3`; on $32^3 \times 64$, for instance, super-blocks 2 and 2).

The spectral quantities (Secs. XII.B–XII.D). The example `preconditioned_fov_device` computes, with exact coarse solves, the field of values, the parts of the cycle, the eigenvalues with their condition numbers, and $\sigma_{\min}(z - \tilde{A}\mathcal{M})$ on contours, e.g.

```
cargo run --release --features cuda-bridge \
    --example preconditioned_fov_device -- \
    16 16 binary conf.bin 1.0 0.001 1 --ratio 10 \
    --arnoldi --far 1,0 --pseudo --contour 1,0,0.6,16
```

(arguments: L, T, format, file, M_5, m, stout $0/1$). It fits 16^4 in 24 GB and takes about an hour there; $24^3 \times 48$ does not fit. The CPU examples `preconditioned_fov`, `coarse_hermiticity` and `correction_exactness` need no GPU; the last, e.g.

```
cargo run --release --example correction_exactness -- \
    12 binary conf.bin 1.0 0.001 1 --block 4
```

prints $\|e(r) - y\|/\|y\|$ for the right form, the left form and the dual correction, and the GCR(2) iterations of $\tilde{A}_R$ and $\tilde{A}_R^\dagger$ (the smoother of setup S by default, `-steps 12 -ratio 30` for setup B; the exactness checks take about 12 minutes on 12^4 on a desktop CPU).

What would help most.

1. The mass scan of Table 10 on three to five further $24^3 \times 48$ configurations, for the spread of the iterations and of the speed-up.
2. The same on a larger volume ($32^3 \times 64$ on an 80 GB GPU), for the volume dependence of the multigrid iterations and of the setup.
3. The mass scan on a dynamical ($2 + 1$-flavor) ensemble, e.g. that of Ref. [24], where the spectra of Sec. III were taken.
4. The spectral run above on further 16^4 configurations, for how the outliers of setup S and the pseudospectra vary between configurations.

For each run, the result lines above and the commit of the build are enough; the logs contain everything else.

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